

AERO70027 – Composite Engineering Practice

Composite Frame Structures for Economy Aircraft Seats

Group 4

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March 23, 2026

1 Summary

This report proposes a composite alternative to aircraft seat frames that are traditionally manufactured using metal alloys. Several concepts were first developed and compared considering cost, manufacturing scalability, assembly, and integration. Materials were selected to find a reasonable combination of performance and weight as well as joining simplicity, and designs were evaluated based on mechanical load thresholds (material failure stress and strain) and relevant failure criteria (Tsai-Hill, Tsai-Wu) in ABAQUS CAE using FEA under static loading conditions derived from the 16g forward load test [1]. The proposed concept is made from both continuous and short fibre reinforced PEEK and has a weight of just 6.2kg, 9% of which is aluminium (base feet). The proposed design is also compared to an alternative design that utilised a sandwich structure bench.

2 Introduction and Problem

Background While manufacturers are able to achieve complete seat assemblies with weights of just 20-30 kg, green aviation initiatives call for further weight reduction, as this can significantly reduce fuel consumption (airline operating cost) and ultimately CO₂ emissions in commercial aircraft. This report proposes a seat frame design whose weight is 90% composite, with a weight of just 6 kg.

Economy aircraft seats are traditionally manufactured with metal alloys; namely aluminium. These alloys are lightweight and can be formed and processed into various profiles due to a combination of low melting temperature and low hardness, which allow them to be manufactured using scalable manufacturing methods, including casting, machining, drawing, and forging. Some of these profiles include extruded tubes and forged geometries, as seen in Fig. 2.1.

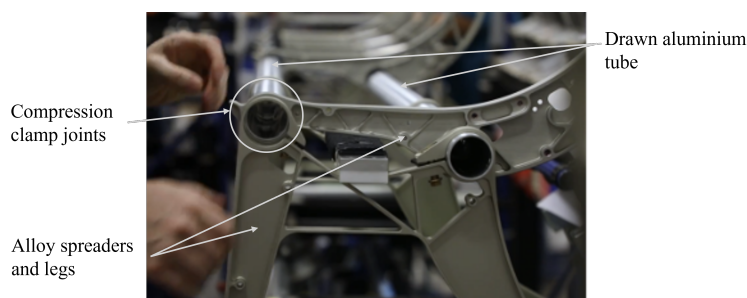


Figure 2.1: Conventional aluminium alloy seat construction [2].

Problems with composite structures design Conventional metal seat constructions are composed of a wide range of components that are assembled using mechanical fasteners. A key design principle in conventional frames is the use of longitudinal tubes, which act as the primary attachment points for spreaders and legs. These components are commonly clamped to the tubes. For metals, tube clamping is a simple and robust assembly method; however, this approach is not suitable for composite structures, since the local compressive loads can cause matrix cracking, crushing, and delamination. Mechanical fasteners also present a major limitation in laminated composites, because they require holes to be drilled through the laminate, which interrupts the load-carrying fibres, creates local stress concentrations, and can lead to bearing failure at stresses much lower to that of the composite's strength [3]. This reduction in joint efficiency is therefore a key driver in the development of the proposed concept. Consequently, alternative joining methods must be considered when designing a seat frame from composite materials.

3 Design philosophy

Identification of primary load paths The seat frame is required to pass the 16g forward load scenario under anti-symmetric passenger occupancy (two passengers, one in centre seat and one on isle/window), effectively simulating a worse case emergency landing. The resultant forces of this impact were presented in 8 loaded nodes and 4 fixed nodes (floor mounting), with effective force magnitudes given in vertical and horizontal directions.

Two studies were performed to evaluate the effective load paths resultant from this loading scenario (modelled in a static model). The first is a generative design study, while the second is a shape optimisation. In both studies, geometric constraints were applied to accommodate passenger leg and baggage room, as well as life jacket compartments. The results can be seen in Fig. 3.1a and Fig. 3.1b. Both studies suggest that the main load path is directed through a simple three-member geometry between the floor mounts and bench.

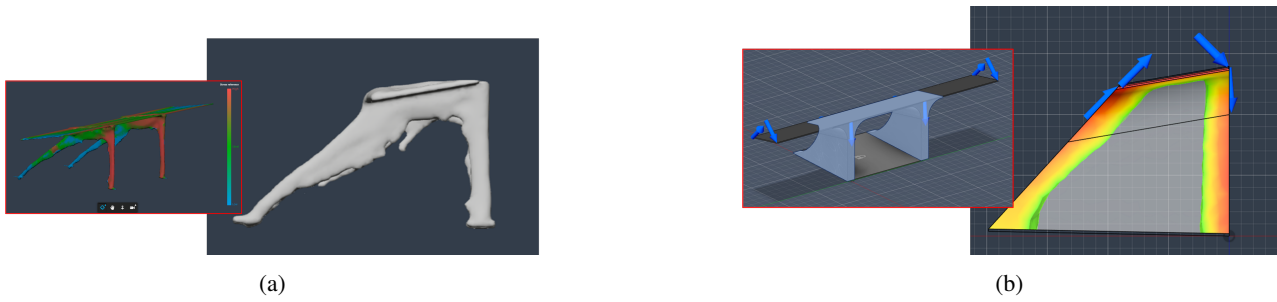


Figure 3.1: Load paths determined by (a) generative design and (b) shape optimisation studies.

Concept development Considering the manufacturing complications introduced by fastener integration, it was decided to design frame structures that do not rely on mechanical fasteners for assembly. The concepts presented were developed to take advantage of existing composite part design practices. These include tubes for lateral and bending load transfer, laminated structures with profiles optimised for high bending stiffness, and adhesive joints. The structures were tested (FEA) with different material combinations to evaluate their impact on weight. This led to the development of designs of which the assembly relied on adhesives and mechanical locking. Initial designs are shown in Fig.3.2. Emphasis was put on short fibre reinforced plastic joints, which can be easily injection moulded as an alternative to joints made from machined aluminium.

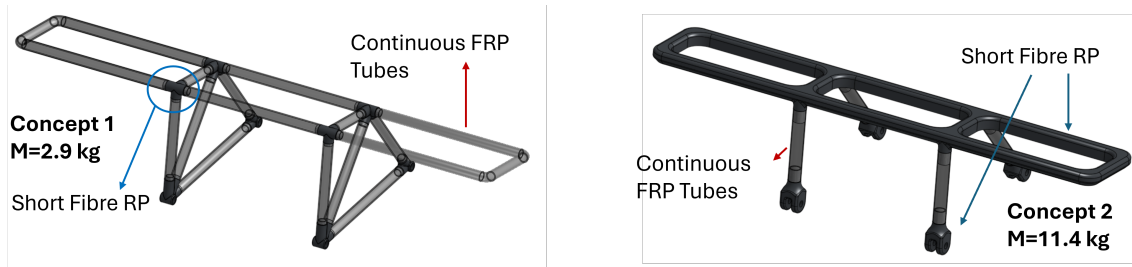


Figure 3.2: Initial concepts utilising simple composite construction.

4 Final frame construction

The frame construction was broken down into two parts - a bench, connecting the loading nodes (nodes 1-8), and the connection to the base feet (nodes 9-12). Initial FEA models showed significant deformation in both concepts shown in Fig.3.2, therefore it was decided to replace the main structure (bench) with a laminated composite shell. The u-section profile of the shell members also allowed for mechanical interlocking to support the alignment of inserts and legs. The final design is shown in Fig.4.1.

The thickness of layups in both the tubes and the shell (bench) were optimised based on stress, deflection, strain, and failure results from FEA simulations, resulting in total thicknesses of 4.48mm for the shell and 3.36mm for the tubes. It was decided to use Aluminium 7075-T6 in the base feet to allow for high torquing to foot mounts on the aircraft. The assembly sequence of the structure can be seen in Fig.4.2. Adhesive details are discussed in Section 6.

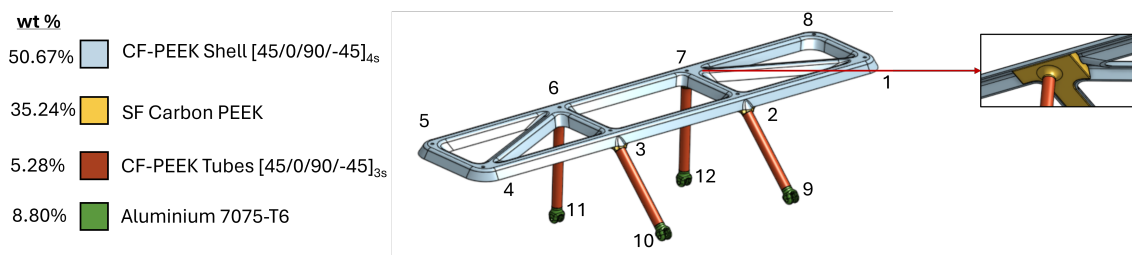


Figure 4.1: Final composite frame structure design. Mass = 6.19 kg, 91% composite weight.

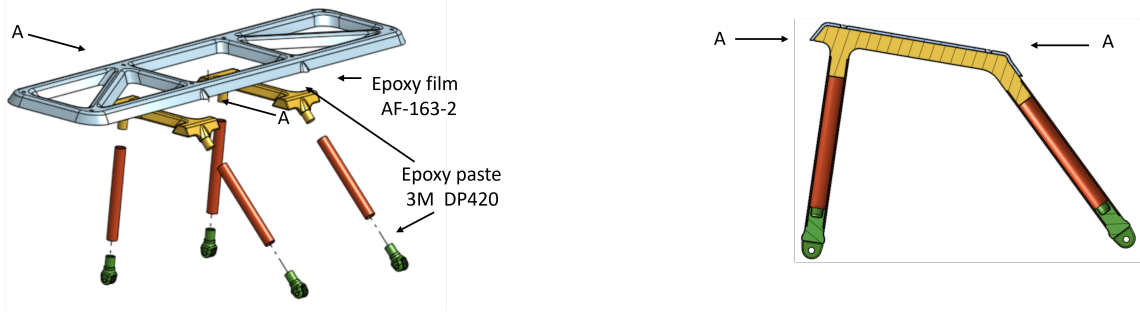


Figure 4.2: Construction assembly and joining.

5 Material selection

5.1 Selection Process

Commercially available thermoplastic prepreg and short fibre composites were evaluated for the composite components of the proposed design. The performance of thermosetting composites was evaluated however despite their superior performance, they were ultimately not considered. Thermoplastic composites can be manufactured at higher production rates and can be reprocessed more easily through processes such as injection moulding and thermoplastic stamp forming [4]. Thermoplastics are also tougher than thermosetting polymers, which can improve resistance to fracture due to sudden seating and foot impact. In this section, Both glass and carbon fibre thermoplastic prepreps were investigated and compared based on specific strength, specific stiffness, cost, and processing temperature. Aramid fibres tend to be commercially unavailable in the desired ranges of thermoplastic matrix prepreps due to their poor interfacial bonding with most polymers [5], hence they were not perused.

5.2 Laminated Composites

In our evaluation, we considered UD prepreps to evaluate materia to their higher fibre volume fraction and allow greater control over fibre orientation, supporting the jig-saw layup preforming strategy used to align fibres with principal load paths. Thermoplastic UD prepreps also offer improved formability compared with thermosets, as elevated temperature reduces matrix stiffness and inter-ply friction, enabling deformation through shear, bending and ply slip mechanisms that reduce wrinkling risk [6]. Experimental results show CF-PEEK formability can increase by approximately 110% at elevated processing temperatures due to increased fibre mobility [7], and post-forming studies confirm the ability to produce curved structural geometries [8].

Table 5.1: Material density and specific mechanical properties for commercially available UD prepreps.

Material Units	Density (g/cm ³)	Specific Mod T (MPa*g/cm ³)	Specific Mod C (MPa*g/cm ³)	Specific Str T (MPa*g/cm ³)	Specific Str C (MPa*g/cm ³)
Toray UD Carbon PEEK (TC1200, AS4D, vf 59.9%, 145gsm)	1.59	84906	77988	1516	818
Toray UD Carbon LMPAEK (TC1225, AS4D, vf 58.6%, 145gsm)	1.59	84906	77988	1516	818
Toray UD Carbon PEKK (TC1320, AS4D, vf N/A%, 145gsm)	1.59	84906	77987	1516	818
Toray UD Carbon PA6 (TC910, N/A, vf 60%, N/Agsm)	1.59	68897	68897	1426	684
Toray UD Carbon PA (TC915, N/A, vf 60%, 131gsm)	1.45	79487	78205	1397	676
Toray UD Carbon PEI (TC1005, T700GC, vf 51%, 120gsm)	1.50	73333	61600	1607	747
Toray UD Carbon PPS (TC1100, AS4A, vf 59%, 221gsm)	1.59	84277	73585	1270	692
TECATEC PAEK GF50 S296 natural	1.9	11053	12632	184	670

Based on the materials shown in Table 5.1, Toray UD Carbon/PEEK (TC1200, AS4D, vf = 59.9, 145 gsm) was selected for the laminated bench structure (Section 4). Although its specific tensile strength is slightly lower than Toray UD Carbon/PEI (TC1005, T700GC, vf = 51, 120 gsm), TC1200 provides higher specific compressive strength, desirable

for the compressive loading in the frame legs. Thermoformed UD thermoplastic composites have been successfully implemented industrially, including thermally press formed tailored blanks for complex bicycle frames by Rein4ced, demonstrating the feasibility of producing complex structural geometries using this approach [9].

Layup selection A symmetric and balanced layup $[45/0/90/-45]_{ns}$ with 3s laminate and 4s laminate stacks was selected for the tubes and the bench structure. They were analysed in Anaglyph Laminate Analysis Program (LAP) with material properties defined with TC1200 and all N_x and N_y values defined as zero (See Fig.B.1 and B.2). This is a thin quasi-isotropic layup that will give the laminates equal stiffnesses E_{xx} and E_{yy} , which is valuable since the loading of the bench and tubes is multiaxial.

5.3 Short Fibre Composites

Table 5.2 shows the material properties of candidate short fibre composites (SFCs). PEEK based composites were prioritised for comparison in order to match that of the laminated composite, for which research has suggested that the adhesion of PEEK-PEEK interfaces can be significantly improved with UV irradiation as discussed in Section 7. Notably, PEEK is one of the few polymers that can maintain its performance after irradiation treatment (LAP shear strength improvement of 350% after 5 second UV irradiation treatment [10].

Table 5.2: Material density and specific mechanical properties for commercially available SFCs.

Material Units	Density (g/cm ³)	Specific Mod T (MPa*g/cm ³)	Specific Mod C (MPa*g/cm ³)	Specific Str T (MPa*g/cm ³)	Specific Str C (MPa*g/cm ³)
Ensinger SF Carbon PEEK (TECAPEEK IM CF30 Black)	1.44	24306	6250	198	198
Hexcel SF Carbon Epoxy (HexMC®-i C/2000/M77-RA)	1.55	24516	24516	194	187
Mitsubishi SF Carbon PEEK (KyronMAX S-9230, vf 30%)	1.41	26241	5674	238	240
Mitsubishi SF Carbon PPA (KyronMAX S-4330, vf 30%)	1.4	29286	6429	257	261

Ensinger SF Carbon PEEK (TECAPEEK IM CF30 Black) was chosen despite its lower specific properties due to its high thermal stability [11], interfacial compatibility [12] and ease of manufacturing [13]. Additionally, PEEK can be reprocessed many times without reducing its properties significantly.

6 Joining and assembly methods

Joining methods were evaluated based on lap shear strength (LSS), manufacturability, and compatibility with aerospace certification requirements (Table 6.1). Joint sizing was calculated using conservative single value LSS data from experimental studies and manufacturer datasheets with a safety factor of 2 to account for uncertainty in bond quality and manufacturing variability, where surface area is formulated by (1).

Table 6.1: Comparison of joining methods for CF-PEEK structural elements

Joining type	Material	Preparation	ILSS (MPa)	Area (mm ²)
Epoxy paste	CF-PEEK → CF-PEEK	Untreated	7.48	13369
Epoxy film	CF-PEEK → CF-PEEK	UV-treated	37.3	2976
Ultrasonic weld	CF-PEEK → CF-PEEK	PEI interleave	28.6	3496
Induction weld	CF-PEEK → CF-PEEK	–	17.6	5682
Plasma weld	CF-PEEK → Al	–	15.5	6452
Laser weld	CF-PEEK → Al	Grit blasted	21.4	4672

$$\text{Surface Area} = \frac{\text{Force}}{\tau_{\text{allow}}/\text{Safety Factor}} \quad (1)$$

Selection PEEK has relatively low surface energy, which reduces adhesion between the substrate and adhesive and therefore limits achievable lap shear strength (LSS). Surface activation methods such as UV irradiation or plasma treatment significantly improve adhesion by increasing surface polarity and wettability, resulting in improved load transfer and a shift from adhesive failure to cohesive or substrate failure. For example, UV irradiation of CF-PEEK bonded using 3M AF-163-2 epoxy film adhesive increased LSS from approximately 8.3 MPa to 37.3 MPa after only 5 seconds of exposure, demonstrating the strong sensitivity of bond performance to surface preparation [10].

Bonding Adhesives formulated for low surface energy (LSE) polymers such as acrylic or urethane may be used however, mechanical testing is required to determine their LSS with CF/PEEK. Toughened epoxy paste and film adhesives are preferred for primary structural joints due to their high fracture toughness and peel resistance. Typical lap shear strengths on aluminium substrates are approximately 31–34 MPa, while untreated CF/PEEK joints may achieve significantly lower values (7–8 MPa), reinforcing the importance of surface preparation.

Composite/Metal Interface Hybrid joining between CF/PEEK and aluminium can be achieved using surface preparation methods such as plasma treatment, grit blasting, or laser texturing. Plasma treated PEEK bonded to aluminium demonstrated LSS values of approximately 15.5 MPa, while optimised aluminium surface preparation achieved values up to 21.4 MPa [14, 15]. Galvanic corrosion between carbon fibre and aluminium can be mitigated by electrically insulating the fibres using a thermoplastic interlayer, such as the PEI film used for UV treatment [16]. Corresponding bonded areas required to transfer a 50 kN load range from approximately 4670–6450 mm², which remains feasible within typical lug geometries.

Selected Joining Method Epoxy adhesive bonding (3M AF163-2) with 5 second UV irradiation surface treatment was selected as the joining method for the bench, and the same preparation for the legs to lugs will use Epoxy paste (DP420). Where mechanical fastening is necessary on the bench to lugs, stress concentrations can be reduced using large diameter washers for the C-channel or inserts to increase bearing area and limit local compressive damage [17].

Overall, automated adhesive bonding with surface preparation is preferred for due to improved stress distribution and compatibility with composite laminates. Thermoplastic welding, although efficient, is more suitable where geometry allows controlled heating and generally results in lower LSS than epoxies [18]. Composite to metal joints remain necessary at the seat track interfaces but require larger bond areas due to lower achievable LSS. Safety margins ensure adequate load transfer despite uncertainty in manufacturing conditions. Mechanical fastening through inserts and large surface clamp interfaces with nuts and bolts would provide adequate redundancy and ease the compliance process for aerospace certification.

7 Structural Analysis (ABAQUS FEA)

A finite element model was developed and simulated for the chosen design. This section outlines the assembly, loading, and result extraction of said model.

7.1 Model Setup

Loading Considerations Loading was to be applied to the model in accordance with the outlines concentrated loads at 8 different node locations on the not surface of the horizontal element of the design. No additional load carried nodes, or the spreading of the required concentrated loads were permitted.

Boundary Conditions With accordance to the brief, the boundary conditions were stated at 4 bottom nodes, where 1 degree of freedom (UR1) was allowed and all the rest were constrained. This boundary conditions were implemented to simulate a roller pin joint, attaching the frame to floor railings.

Materials and laminate assignment Material properties were assigned to every part within the imported CAD assembly, using either a homogenous section assignment or through a composite laminate layup tool. The C-channel bench alongside the tubes (after being turned into shell elements) were assigned composite layups using the chosen UD Carbon PEEK properties, whilst the base feet and inserts were both assigned solid homogenous sections, using the Aluminium 7075 and BMC Carbon PEEK respectively. In laminate parts, through thickness (normal) and 0° fibre directions were set across the surfaces of the corresponding laminate parts and a nominal thickness of 0.14mm was assigned to each ply within the stacking sequence. The plies were checked for symmetry, making them generate equally from both sides of the shell surface.

7.2 Assembly of Instances

After successful section assignment of all parts within the model, partitions were made for all composite layup parts in order to simulate the relevant adhesion surfaces. Tubes were given partitions 40mm away from each end (being of equal

length to the fitment lips found on the inserts and base feet), whilst more complex partitions were made on the bench part. These composite partition faces, as well as the corresponding faces of the inserts and base feet to be joined to, were then assigned surfaces, with special care going into choosing the correct side of the composite elements that were to be realistically used for adhesion. Tie constraints were chosen to assemble the parts, with the constraints being applied only to the chosen partitioned adhesion regions. Fig.7.1a and Fig.7.1b illustrate the bench-insert tie constraint and tube-base feet tie constraint respectively.

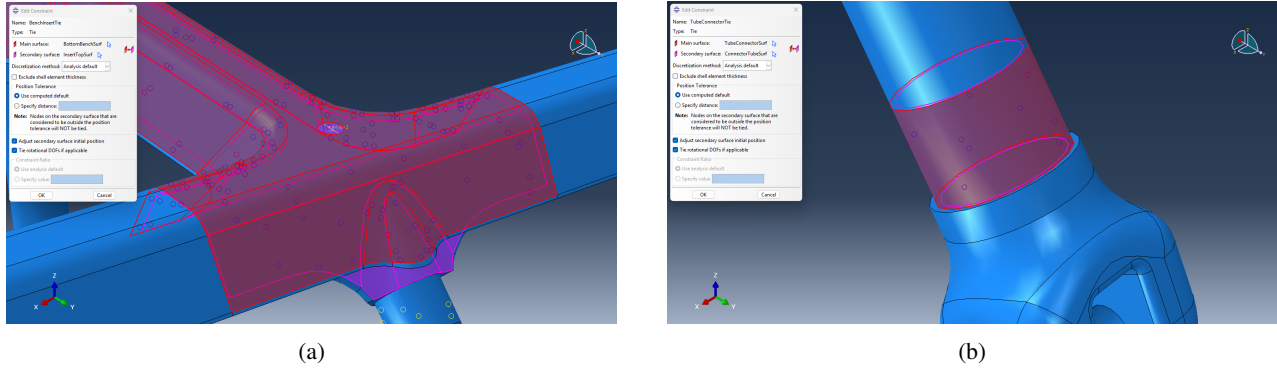


Figure 7.1: Tie Constraints. Insert-tie constraint (a) and tube-base feet tie constraint (b).

7.3 Loading Assignment and Boundary Conditions

12 reference points were created and placed at the required node coordinates, with 8 top nodes for loading and 4 bottom nodes for the floor constraints. The acting concentrated loads were assigned to the top nodes at the centres of the joining apertures and hence had to be distributed to the existing geometry through the use of coupling constraints. In a similar fashion, the bottom, constrained nodes were coupled to the inner surface of the adapter through holes, in order to simulate the constraints of the roller pins. Fig.C.1 and Fig.C.2 illustrate the loading node coupling and boundary node coupling respectively. Additionally, Fig.C.3 and Fig.C.4 in Appendix C show the final load and boundary condition allocations respectively.

Mesh Assignment Mesh refinement was attempted, however due to the Educational License of ABAQUS being the only version available, the number of nodes was restricted to 250,000. Lower resolution were tested however results suggested that elements resolution was too small. A combination of quad-biased and tetrahedral element types were used across the assembly geometry, ranging from 2-6mm.

Job Creation and Field Outputs In total, 4 jobs were created to fully analyse the assembly and all instances within it. 3 laminate analysis jobs were run on the bench, front legs and rear legs, which provided results for principal/shear stresses, strains and Tsai Hill failure criteria of all the distinct plies. A final, whole model, job was run in order to provide stress and strain distributions of the homogenous sections, as well as to display the deformation displacements of the frame assembly.

7.4 CFRP vs KEVLAR® laminate performance

As part of the material selection process, CFRP was compared to KEVLAR® in the laminate material assignment. Primarily, KEVLAR® was chosen as the comparative material due to its reduced density compared to CFRP (15%). However, post simulation strain and failure criteria results were not satisfactory, and failing against the specified requirements.

To satisfy these requirements, the KEVLAR® structure would have needed to be approximately 30% heavier than the CFRP alternative and was therefore not pursued as the preferred material. Table 7.1 shows comparative result values of CFRP vs KEVLAR® extracted from their respective whole model simulations using equivalent laminate thickness, highlighting the rationale behind the material selection decision.

Table 7.1: Comparison of CFRP and Kevlar composite results. MAX properties shown. E = Strain, U = deflection.

	S. Mises [MPa]	E [%]	U [mm]	S. Principal [MPa]				Tsai-Hill Criterion			
				-45° Ply	90° Ply	0° Ply	-45° Ply	-45° Ply	90° Ply	0° Ply	-45° Ply
CFRP	792	0.984	43.1	771	652	1301	659	0.724	1.063	0.579	0.696
KEVLAR®	723	2.07	83.3	702	603	1209	593	3.246	5.319	2.539	3.116

7.5 Alternative designs

A competing concept was developed consisting of a sandwich structure in place of the C-channel bench. The model setup was identical to the one outlined in the above subsections, with the exception of new foam properties and section being assigned to the core structure.

A comparison of basic results (stress, strain, displacement) seen in Table 7.2, clearly demonstrating the rationale behind the selection of the chosen design. Although the sandwich structure concept achieved a 15% weight reduction, the superior performance the selected design was the decisive factor in its selection. It should be noted that further optimisation of the sandwich concept is required to create a more accurate comparison.

Table 7.2: Comparison of the chosen design and sandwich concept. MAX properties shown. E = strain, U = deflection.

	S. Mises [MPa]	E [%]	U [mm]	S. Principal [MPa]				Tsai-Hill Criterion			
				45° Ply	90° Ply	0° Ply	-45° Ply	45° Ply	90° Ply	0° Ply	-45° Ply
Proposed	792	0.984	43.1	771	652	1301	659	0.724	1.063	0.579	0.696
Sandwich st.	8000	7.39	73.73	7666	7863	7092	6033	7.16	6.12	5.99	4.74

7.6 Final design performance and future optimisation

A safety factor of 1.5 was applied to the specified baseline requirements (stress, strain, displacement, deflection angle) and these adjusted values were used to determine the validity of the design and simulation. The safety factor thresholds were derived from both the material failure properties and relevant regulatory standards. The Tsai-Hill failure criterion was selected to evaluate the structural integrity of each ply at discrete orientations. Table 7.3 outlines the simulation results in comparison with the safety margin requirements.

As seen in Table 7.3, although as a whole the concept was able to satisfy most of the performance criteria, the inserts were decided to require further optimisation. The main cause of failure seen in the inserts was thought to come from the stress concentrations built up at the sharp transition around the corners (seen in Fig.D.6). Further optimisation of geometry and material selection is needed to increase the structural conformity of said insert parts.

Additionally, the table indicates that the 90° ply in the bench component marginally fails the Tsai-Hill criterion, and hence further layup optimisation is advised. Detailed plots and contours of the mentioned results are provided for reference in Appendix D and E.

To validate the adhesive used, the shear strength (S13), which characterises the lap shear between bonded interfaces (e.g. insert-tube and insert-bench), was evaluated. Contour plots of the insert component, as seen in Fig.D.8, suggest that the adhesive may not maintain lap load transfer uniformly (at a 1.5 safety factor) across the full bond interface. Consequently, the introduction of a secondary mechanical fastener is proposed to increase conservatism and enhance the robustness of the joint.

Table 7.3: Ply-level laminate results for the bench, front leg, and rear leg.

Component / Ply	Max E [%]	Max In-Plane Tensile S. Principal [MPa]	Max In-Plane Compressive S. Principal [MPa]	Tsai-Hill
Bench 45° Ply	0.98	771	647	0.724
Bench 90° Ply	0.97	652	438	1.063
Bench 0° Ply	0.96	1301	753	0.879
Bench -45° Ply	0.95	659	600	0.696
Front leg 45° Ply	0.17	-	32.1	0.482
Front leg 90° Ply	0.17	-	789	0.608
Front leg 0° Ply	0.17	212	18	0.617
Front leg -45° Ply	0.17	-	29.8	0.463
Rear leg 45° Ply	0.61	488	172	0.468
Rear leg 90° Ply	0.59	781	486	0.384
Rear leg 0° Ply	0.56	80	180	0.617
Rear leg -45° Ply	0.53	308	297	0.460
Threshold	1	1606	866.7	1

Table 7.4: Local stress and strain results for the insert and base feet.

Component	Max Tensile S. Principal [MPa]	Max Compressive S. Principal [MPa]	Max E [%]
Insert	368	738	1.72
Safety factor requirement	190	190	0.53
Base feet	244	266	0.282
Safety factor requirement	335	335	7.3

Table 7.5: Global response results for the whole model.

Model	Max U [mm]	Angle [°]
Whole model	43.1	8.4
Safety factor requirement	-	13.3

8 Manufacturability assessment and weight-cost evaluation

8.1 Manufacturability Assessment

Optimisation for manufacturing was a key consideration from the first concept design. The design focused on discovering processes that could achieve flexibility in layup, automation and production capacity needed. After evaluating all the suppliers, it was concluded that pull-winding by Exel Composites Ltd [19], injection moulding via the ENGEL Victory 180 [20], and thermoplastic stamp forming assisted with jig-saw preform approach by Pinette PEI would be the best solutions for the tubes, insertions, and top panel respectively [21]. Thermoplastic stamp forming is an approach currently increasingly popular and research is on the applicability of both UD and woven fabrics is being actively developed [22].

8.2 Weight-Cost Evaluation

Using the standard cost model, as defined by Ashby et al, we were able to estimate the cost of manufacturing each part [23]. This analysis was assisted with the cost modelling tool in Granta 2022 R2, which allowed us to estimate the cost per unit (C_s) with the assumptions and input values as detailed in Appendix A.2. A total cost of £1085.20 was estimated for the CFRP frame developed, over two-times higher than a standard aluminium frame, which was estimated at £492.40. To determine the financial feasibility, we accounted for the cost of flying per kg (weight-cost) using a case study, as detailed in Appendix A.3. Using British Airway's London Heathrow-Barcelona round trip served by four A320neos, we investigated when the breakeven point would be if the aluminium frames were replaced with CFRP frames. This was found to be 1.84 years, as seen in Figure 8.1.

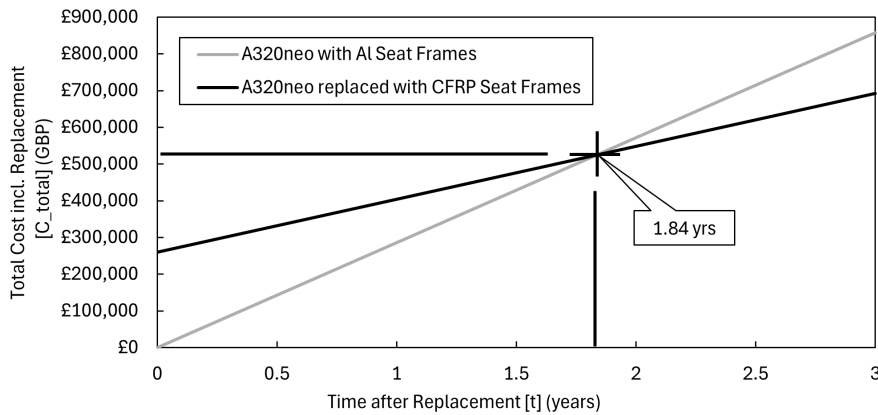


Figure 8.1: Return on investment for replacing Al frames with CFRP frame was conducted for London-Barcelona round trip for four Airbus A320neo with 60 frames (180 seats).

9 Conclusions

In conclusion, this report details the design and simulation of a 6.19 kg aircraft seat frame designed to the requirement of 90% composite materials. Made from a fully carbon-PEEK construction of Toray TC1200 and Ensinger TECAPEEK IM CF30 and joined using 3M AF163-2 with UV irradiation surface treatment, this CFRP frame was proven to meet the required loads expected under a 16G forward load test and was shown to be manufacturable with significant commercial value. Potential improvements for this proposal include further investigation of replacing the outer layers using woven plies for drapability and improved crack resistance. Simulations on reduction in matrix fraction needs to also be further investigated which was not accounted for in the simulation. Experimental work to better inform the design developed and verified numerically in ABAQUS is also necessary to be able to verify the design.

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Appendices

A Materials and Costing

A.1 Initial Selection

Material selection was conducted by evaluating all PMCs using Ansys® Granta Edupack, Release 2022 R2 (Ansys Inc, USA). The analysis was conducted using a custom-defined library where MMC or CMC were not considered. This was subject to increased complexity in processing, operating conditions not requiring extreme temperatures, and extensive costs.

Upon completion of the filtering stage, to select the best material for each application, the material parameters were isolated for tensile/compression (E/ρ) and bending ($E^{1/3}/\rho$). By plotting the axial tensile strength, and axial compression strength against density, the highest performing materials were isolated as seen in Figure A.1.

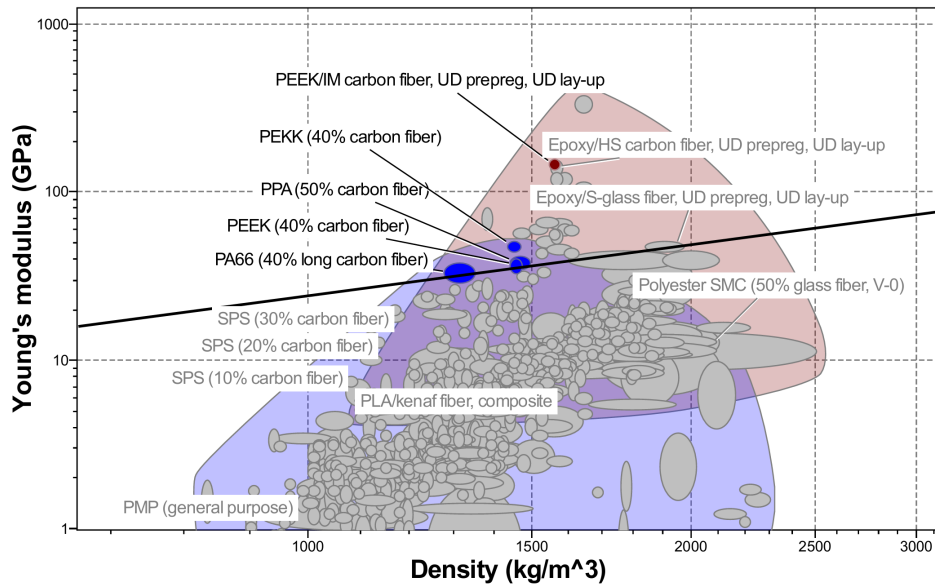


Figure A.1: Density selection map with gradient lines plotted for optimising to tensile and bending loads. Polyamide (PA66), polyetheretherketone (PEEK), polyphthalamide (PPA), and polyetherketoneketone (PEKK) matrix systems were found to be most optimal.

A.2 Material Costing

The total cost of production includes material, capital, time, energy, and any royalty payments and were defined using material cost including scrap rate (C_1), tooling cost (C_2), capital cost of which accounts for productivity (C_3), and overhead costs (C_4) which accounts for labour. Combined, we can define the function as:

$$C_s = C_{\text{material}} + \frac{C_{\text{dedicated}}}{n} + \frac{\dot{C}_{\text{capital}} + \dot{C}_{\text{overhead}}}{\dot{n}} = \frac{mC_m}{1-f} + \frac{C_t}{n} \left[\text{Int} \left(\frac{n}{n_t} + 0.51 \right) \right] + \frac{1}{\dot{n}} \left(\frac{C_c}{L_{\text{two}}} \right) (1+d)^{t_{\text{wo}}} + \frac{\dot{C}_{oh}}{\dot{n}} \quad (2)$$

Key assumptions made included exclusion were to exclude capital costs (machinery) are not included due to the capital write off laws. This allows direct comparison with conventional aluminium products. Profit margins (mark up) and upholstery are considered the same. Cost estimation includes frame cost only. The input values are summarised in the table below.

Table A.1: Assembly cost input variables

Variable	Value	Unit
Mass	6.35	kg
Assembly labour cost per hour	50	GBP/hr
Technical labour cost per hour	113	GBP/hr
Number of assembly lines	1	line(s)
Number of employees on assembly line	20	pax
Annual production	36750	assemblies
Production rate	4	assemblies per hour (24/7)
Cost per assembly	250	GBP/hr per assembly

Table A.2: Raw material cost breakdown for CFRP frame.

Component	Specification	No. components	Mass/component (kg)	Material Price (GBP/kg)	Raw Part Price (GBP/part)	Total Raw Price (GBP)
CF/PEEK Top Plate	Toray UD Carbon PEEK (TC1200, AS4D, vf 59.9%, 145gsm)	1	3.218	90.00	289.58	289.58
BMC CF/PEEK Inserts	Ensinger BMC Carbon PEEK (TECAPEEK IM CF30 Black)	2	1.119	69.00	77.20	154.40
CF/PEEK Tubes	Toray UD Carbon PEEK (TC1200, AS4D, vf 59.9%, 145gsm)	4	0.084	90.00	7.54	30.18
Aluminium Lugs	Aluminium 7XXX ASM Al 7075-T6	4	0.140	3.50	0.49	1.96
Assembly Cost (GBP)						0.00
Total (GBP)						476.11

Table A.3: Process cost breakdown for CFRP frame.

Component	Process	No. components	Process Part Price (GBP/part)	Total Process Price (GBP)
CF/PEEK Top Plate	Thermoforming and CNC milled for precision	1	456.00	456.00
BMC CF/PEEK Inserts	High-pressure injection moulding and CNC milled for precision	2	109.00	218.00
CF/PEEK Tubes	Pullwinding, followed by cutting and trimming process	4	19.30	77.20
Aluminium Lugs	Forged and CNC milled for precision	4	21.00	84.00
Assembly Cost (GBP)				250.00
Total (GBP)				1085.20

Table A.4: Raw material cost breakdown for standard aluminium frame, modified from [24].

Material	Specification	No. components	Mass per component (kg)	Material Price (GBP/kg)	Raw Part Price (GBP/part)	Total Raw Price (GBP)
Aluminium Bending Stiffeners Bars	Aluminium 2XXX Al 2024-T6	2	1.00	1.88	1.88	1.88
Aluminium Frame Legs	Aluminium 7XXX Al 7075-T6	2	2.31	3.50	8.07	18.60
Aluminium Spreaders	Aluminium 7XXX Al 7075-T6	4	1.00	3.50	3.50	3.50
Assembly Cost (GBP)						0.00
Total (GBP)						23.98

Table A.5: Process cost breakdown for standard aluminium frame, values extracted from Granta 2022 R2.

Material	Process	No. components	Process Part Price (GBP/part)	Total Process Price (GBP)
Aluminium Bending Stiffeners Bars	Extrusion and anodised	2	28.90	57.80
Aluminium Frame Legs	Forged and anodised	2	37.50	75.00
Aluminium Spreaders	Forged and anodised	4	27.40	109.60
Assembly Cost (GBP)				250.00
Total (GBP)				492.40

A.3 Business Case Study

Includes indirect labour, facility cost, equipment, and consumables. Since each direction there are 8 flights, therefore for 16 flights total, including turnaround time of 45 minutes, we can conclude 4 aircraft required to service this route.

Table A.6: Aircraft seat cost and fuel saving analysis

Variable	Value	Unit
Composite Aircraft Seat Mass	6.349365	kg
Composite Aircraft Seat Price	1085.20	GBP
Aluminium Aircraft Seat Mass	12.61	kg
Aluminium Aircraft Seat Price	492.40	GBP
Weight saved per assembly	6.260635	kg
Assemblies in Aircraft	60	assemblies
Weight saved per aircraft	375.6381	kg
Cost of fuel	1.1	GBP/kg fuel per kg carried
Fuel burned per kg per hour	0.03	–
Flight hours	4	hrs
Number of flights annually	2860	flights
Flight hours total	11440	hrs
Composite seat fuel cost	12.5717427	GBP/hr
Composite seat fuel cost annual	143820.736	GBP
Aluminium seat fuel cost	24.9678	GBP/hr
Aluminium seat fuel cost annual	285631.632	GBP
Fuel cost saved per hr	12.3960573	GBP/hr
Fuel cost saved annual	141810.896	GBP
Number of aircraft required for round trip	4	aircraft
Total cost for replacement	260448	GBP
Breakeven point	1.83658667	years
Total cost at breakeven	524587.247	GBP

B Manufacturing

B.1 Layup Procedure

Using the [45/0/90/-45] laminate stack and using the material properties of Toray CF/PEEK TC1200, we were able to calculate the following ABD matrix for the laminate stack.

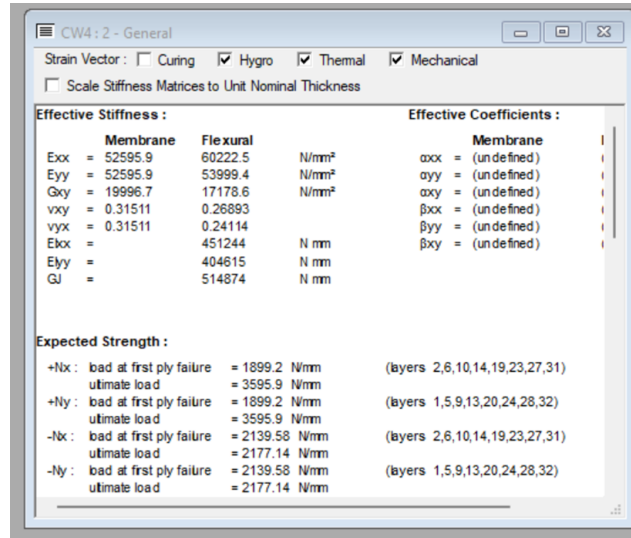


Figure B.1: Effective stiffness calculated from LAP

$$[ABD] = \begin{bmatrix} 261606 & 82435.4 & 0 & 0 & 0 & 0 \\ 82435.4 & 261606 & 0 & 0 & 0 & 0 \\ 0 & 0 & 89585.4 & 0 & 0 & 0 \\ 0 & 0 & 0 & 483265 & 117088 & 9695.53 \\ 0 & 0 & 0 & 117088 & 433402 & 9695.53 \\ 0 & 0 & 0 & 9695.53 & 9695.53 & 129046 \end{bmatrix}$$

Figure B.2: ABD matrix calculated from LAP

Table B.1: Properties for symmetric balanced laminates

Homogenized Property	Symmetric Balanced Laminate
ν_{xy}	$\frac{A_{12}}{A_{22}}$
ν_{yx}	$\frac{A_{12}}{A_{11}}$
E_x	$1 - \frac{A_{12}^2}{A_{22}A_{11}} \frac{A_{11}}{h}$
E_y	$1 - \frac{A_{12}^2}{A_{22}A_{11}} \frac{A_{22}}{h}$
G_{xy}	$\frac{A_{66}}{h}$

C FEA setup

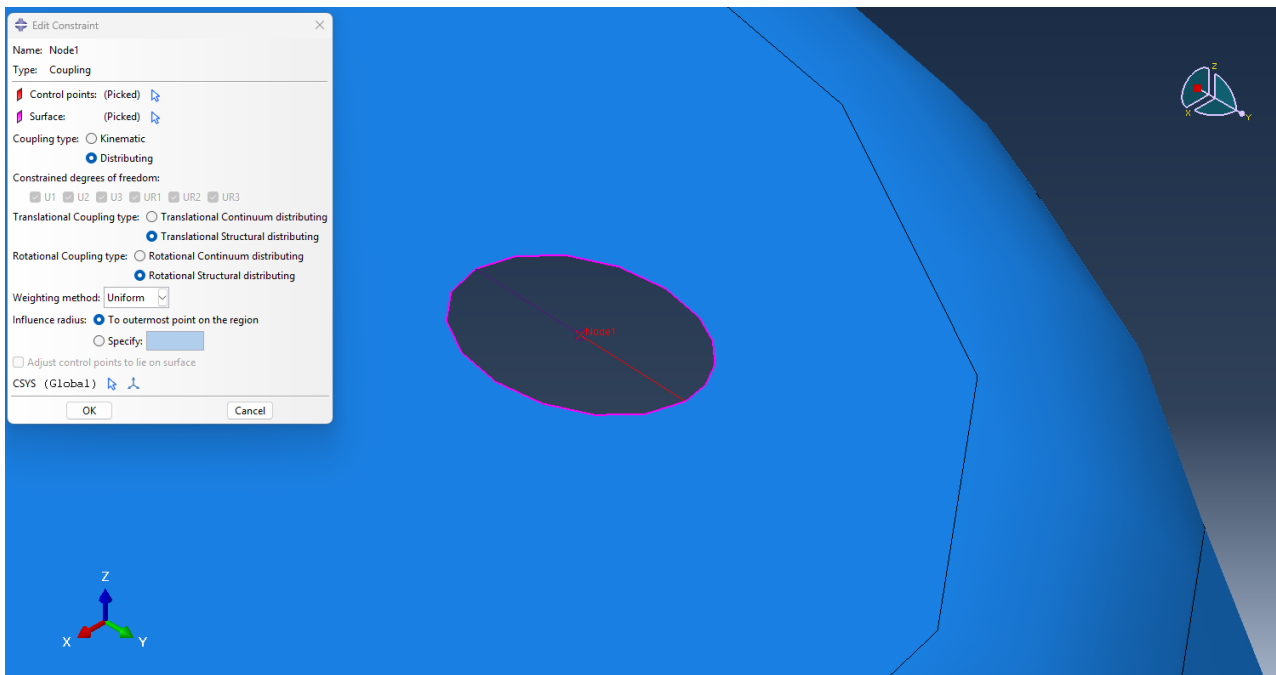


Figure C.1: Loading node coupling

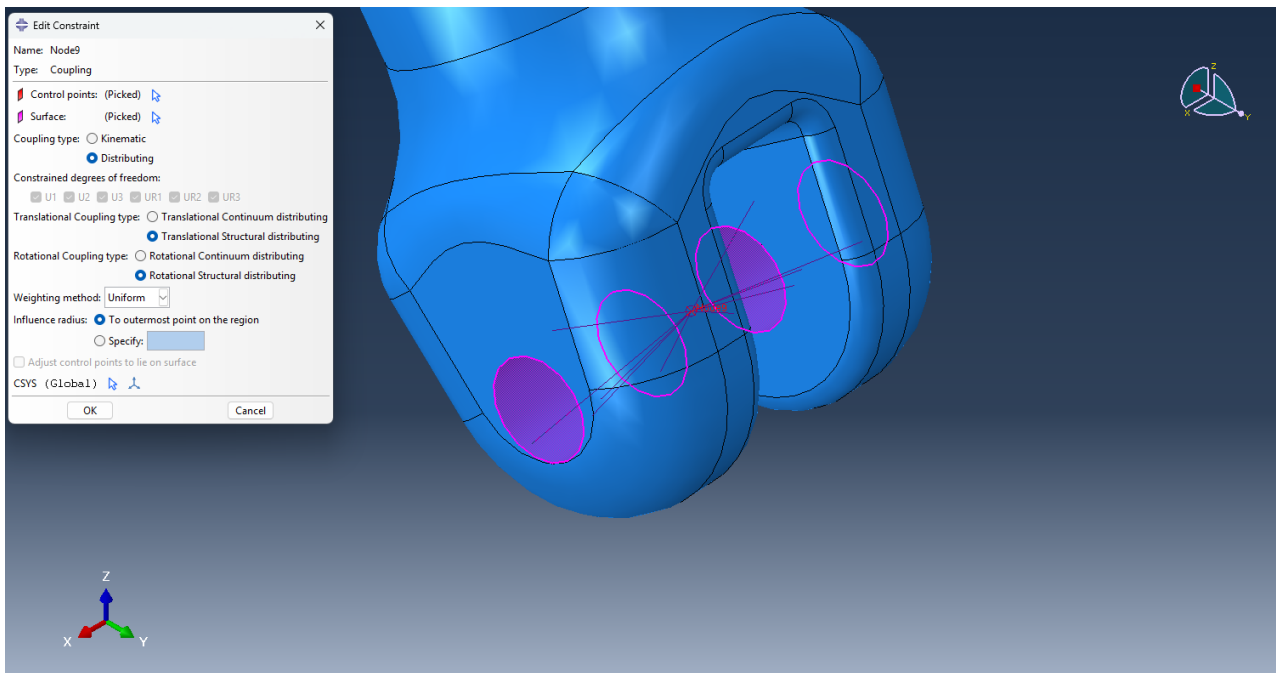


Figure C.2: Boundary constraint coupling

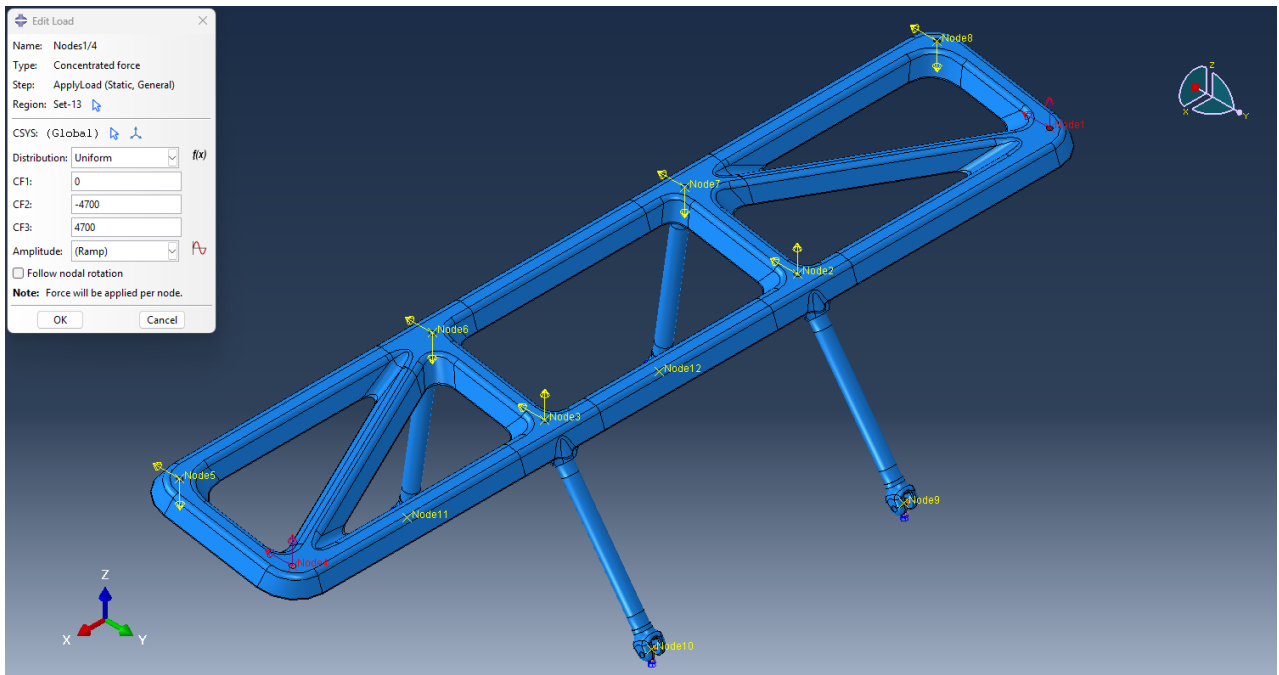


Figure C.3: Loading node assignment

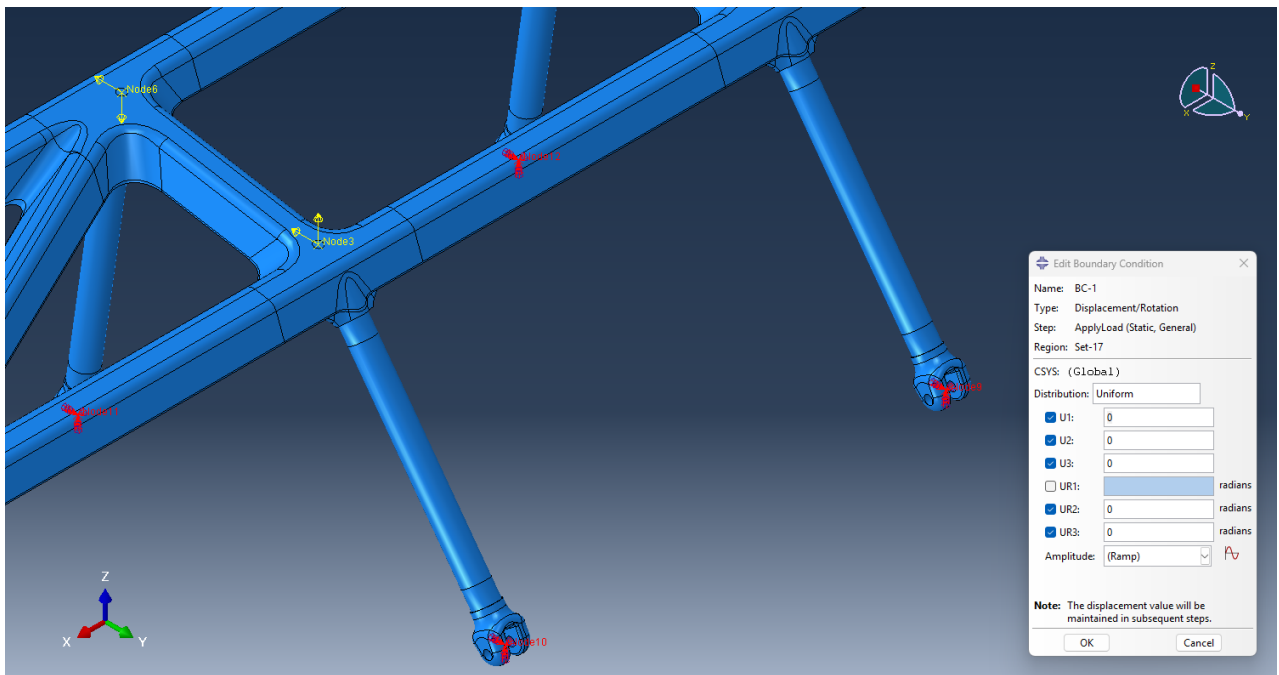


Figure C.4: Boundary Condition node assignment

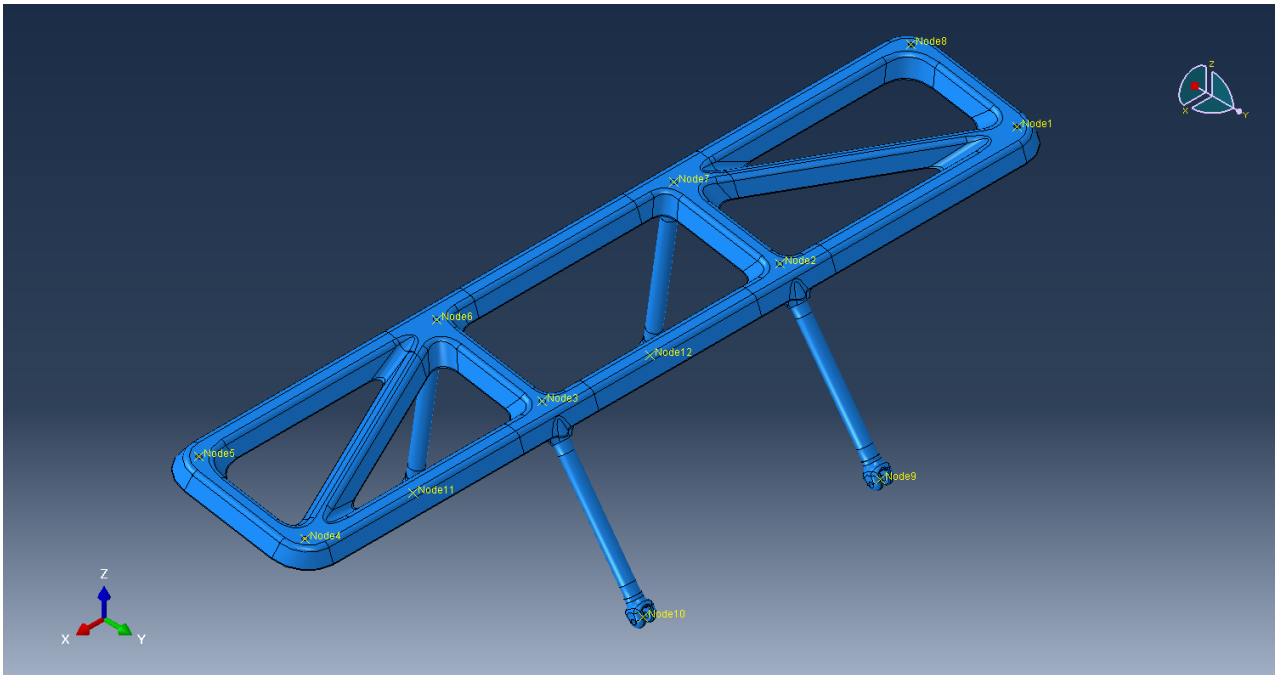


Figure C.5: Chosen design assembled geometry

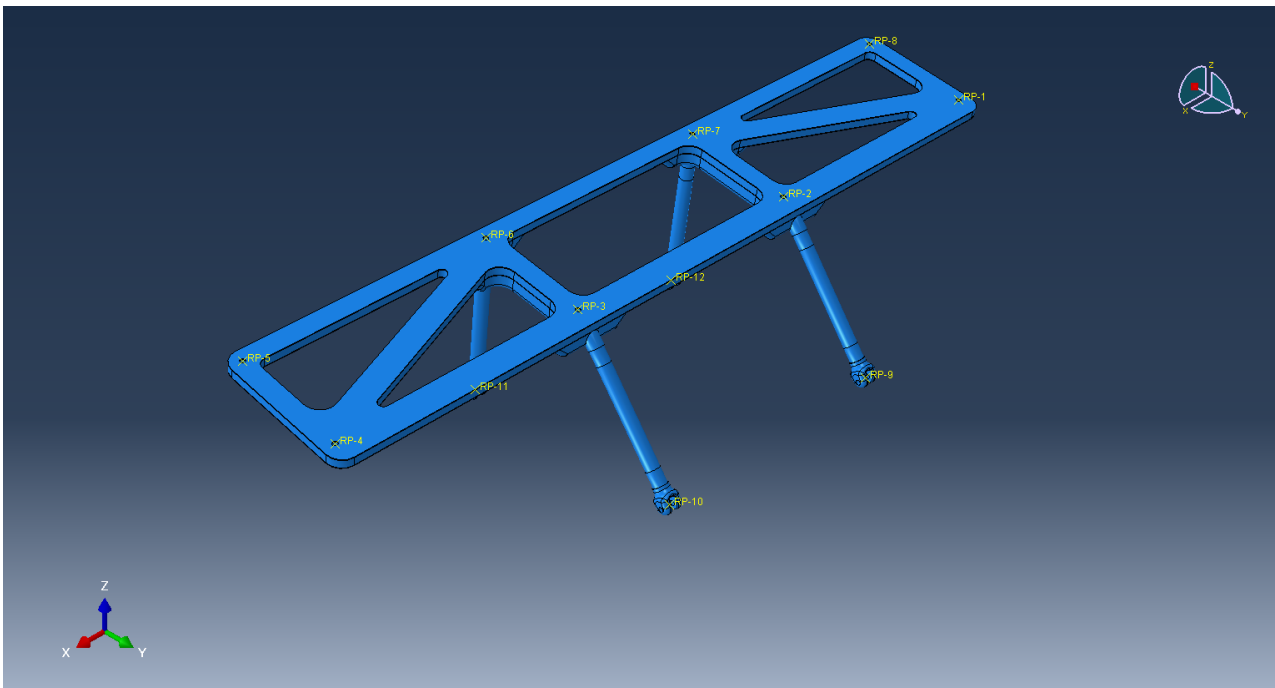


Figure C.6: Sandwich concept assembled geometry

D Homogenous section results

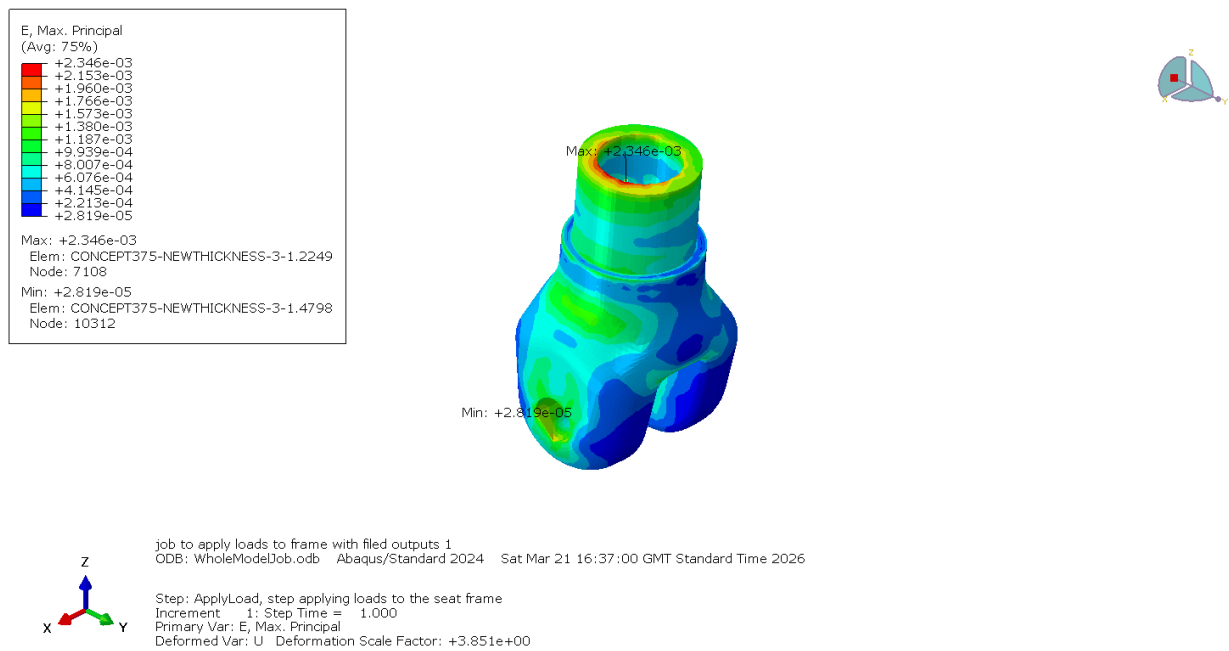


Figure D.1: Front base foot strain contour

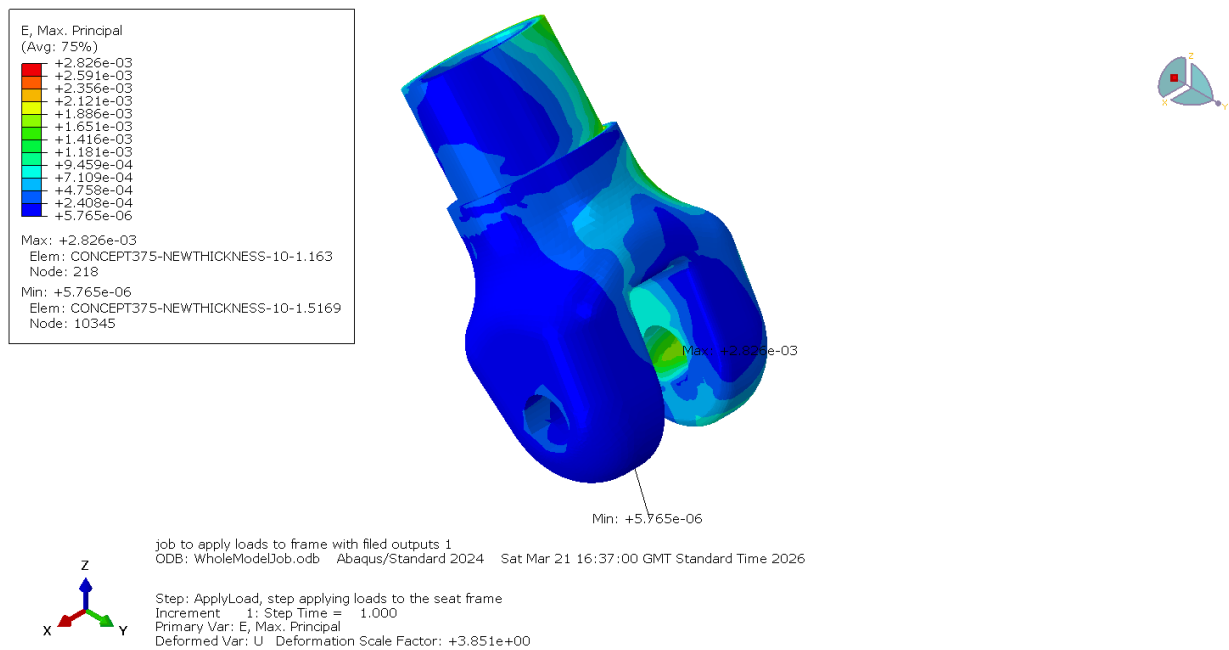
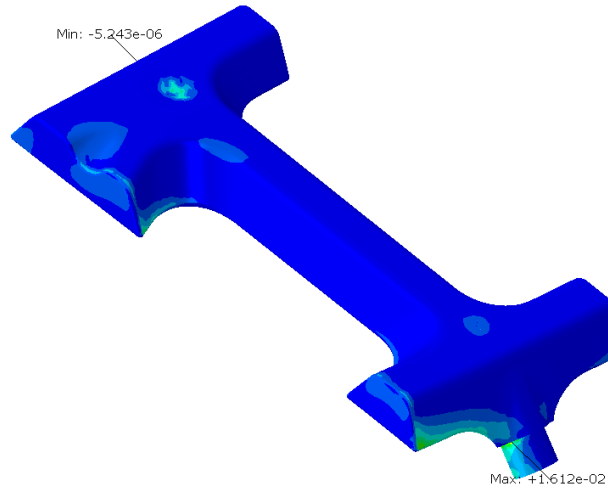
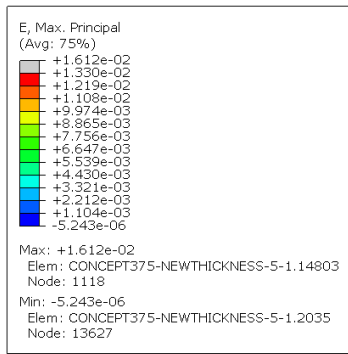


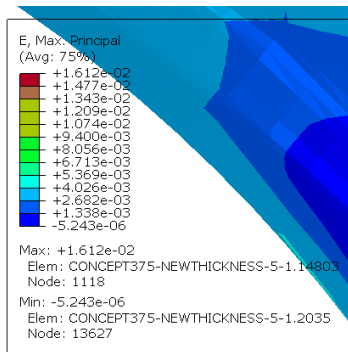
Figure D.2: Rear base foot strain contour



job to apply loads to frame with filed outputs 1
ODB: WholeModelJob.odb Abaqus/Standard 2024 Sat Mar 21 16:37:00 GMT Standard Time 2026

Step: ApplyLoad, step applying loads to the seat frame
Increment 1: Step Time = 1.000
Primary Var: E, Max. Principal
Deformed Var: U Deformation Scale Factor: +3.851e+00

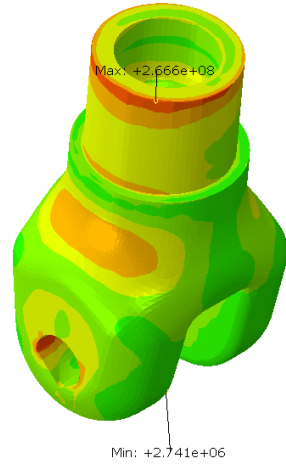
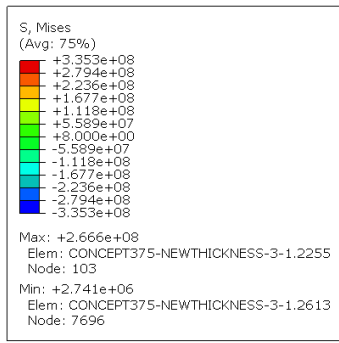
Figure D.3: Insert strain contour



job to apply loads to frame with filed outputs 1
ODB: WholeModelJob.odb Abaqus/Standard 2024 Sat Mar 21 16:37:00 GMT Standard Time 2026

Step: ApplyLoad, step applying loads to the seat frame
Increment 1: Step Time = 1.000
Primary Var: E, Max. Principal
Deformed Var: U Deformation Scale Factor: +3.851e+00

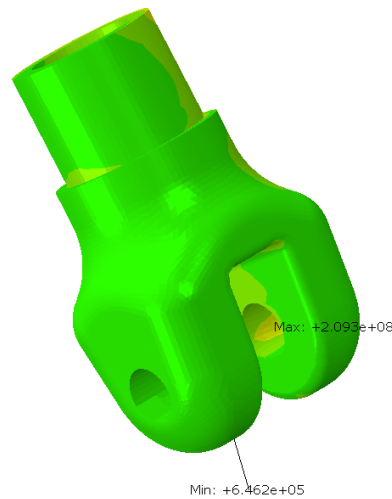
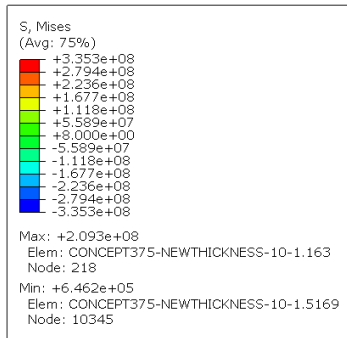
Figure D.4: Insert strain contour zoom



job to apply loads to frame with filed outputs 1
ODB: WholeModelJob.odb Abaqus/Standard 2024 Sat Mar 21 16:37:00 GMT Standard Time 2026

Step: ApplyLoad, step applying loads to the seat frame
Increment 1: Step Time = 1.000
Primary Var: S_y Mises
Deformed Var: U Deformation Scale Factor: +3.851e+00

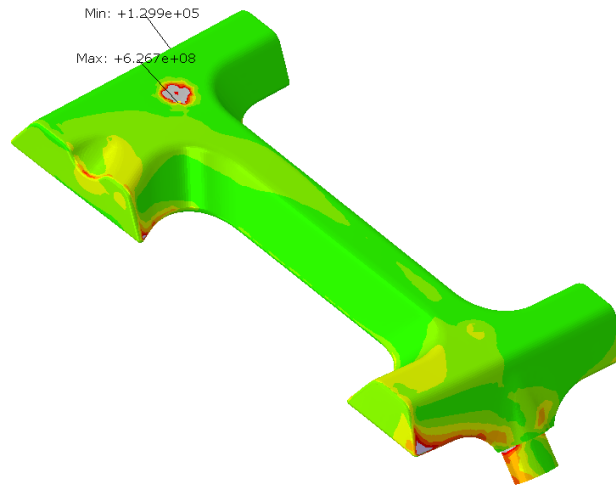
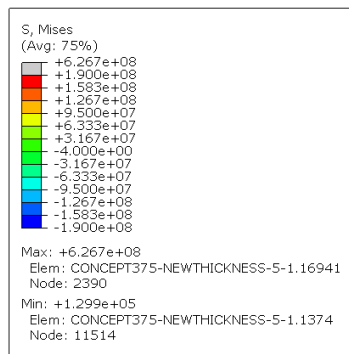
Figure D.5: Front base foot stress contour



job to apply loads to frame with filed outputs 1
ODB: WholeModelJob.odb Abaqus/Standard 2024 Sat Mar 21 16:37:00 GMT Standard Time 2026

Step: ApplyLoad, step applying loads to the seat frame
Increment 1: Step Time = 1.000
Primary Var: S_y Mises
Deformed Var: U Deformation Scale Factor: +3.851e+00

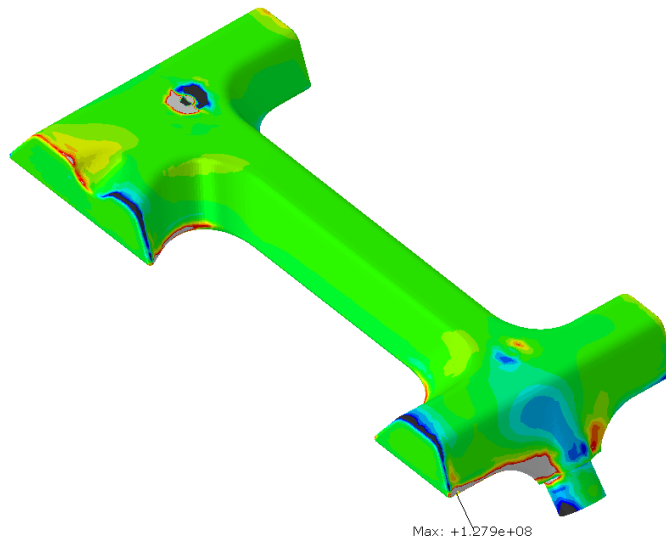
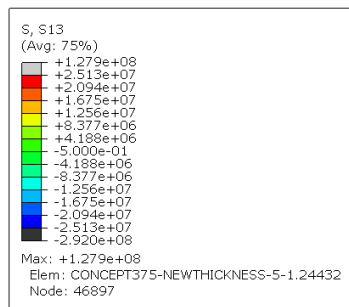
Figure D.6: Rear base foot stress contour



job to apply loads to frame with filed outputs 1
ODB: WholeModelJob.odb Abaqus/Standard 2024 Sat Mar 21 16:37:00 GMT Standard Time 2026

Step: ApplyLoad, step applying loads to the seat frame
Increment 1: Step Time = 1.000
Primary Var: S_y Mises
Deformed Var: U Deformation Scale Factor: +3.851e+00

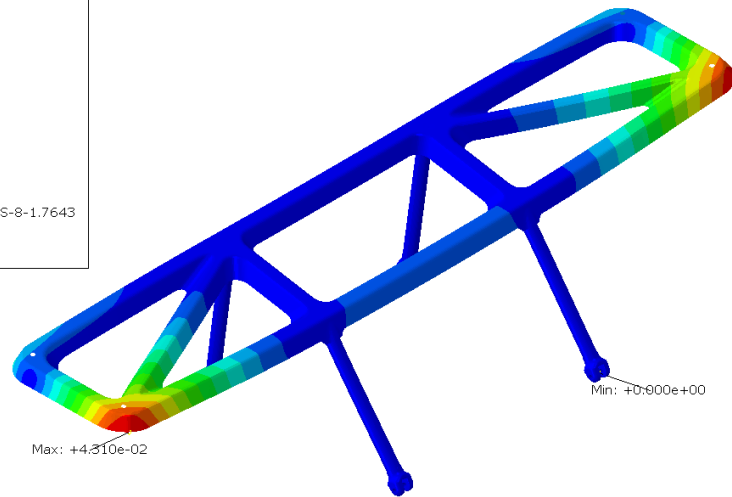
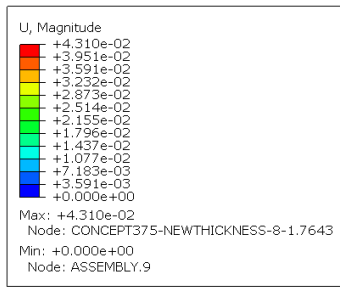
Figure D.7: Insert stress contour



job to apply loads to frame with filed outputs 1
ODB: WholeModelJob.odb Abaqus/Standard 2024 Sat Mar 21 16:37:00 GMT Standard Time 2026

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Increment 1: Step Time = 1.000
Primary Var: S_y S13
Deformed Var: U Deformation Scale Factor: +1.000e+00

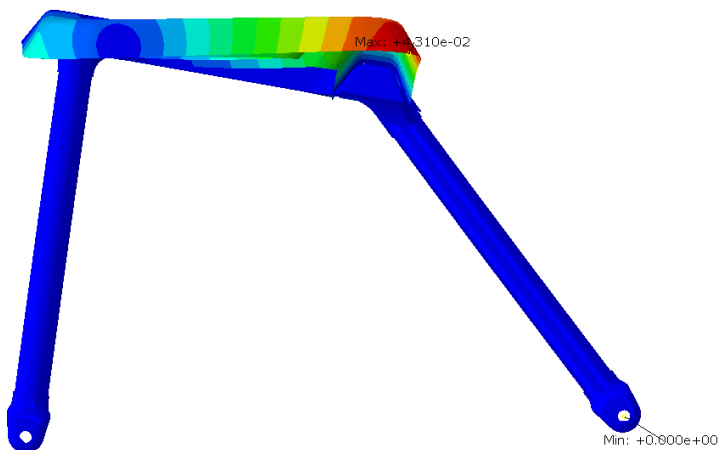
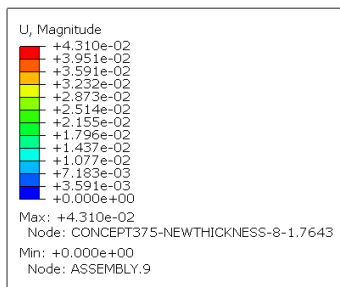
Figure D.8: Insert lap shear stress contour



job to apply loads to frame with filed outputs 1
ODB: WholeModelJob.odb Abaqus/Standard 2024 Sat Mar 21 16:37:00 GMT Standard Time 2026

Step: ApplyLoad, step applying loads to the seat frame
Increment 1: Step Time = 1.000
Primary Var: U, Magnitude
Deformed Var: U Deformation Scale Factor: +1.000e+00

Figure D.9: Whole model deflection in ISO view



job to apply loads to frame with filed outputs 1
ODB: WholeModelJob.odb Abaqus/Standard 2024 Sat Mar 21 16:37:00 GMT Standard Time 2026

Step: ApplyLoad, step applying loads to the seat frame
Increment 1: Step Time = 1.000
Primary Var: U, Magnitude
Deformed Var: U Deformation Scale Factor: +1.000e+00

Figure D.10: Whole model deflection in Left view

E Laminate section results

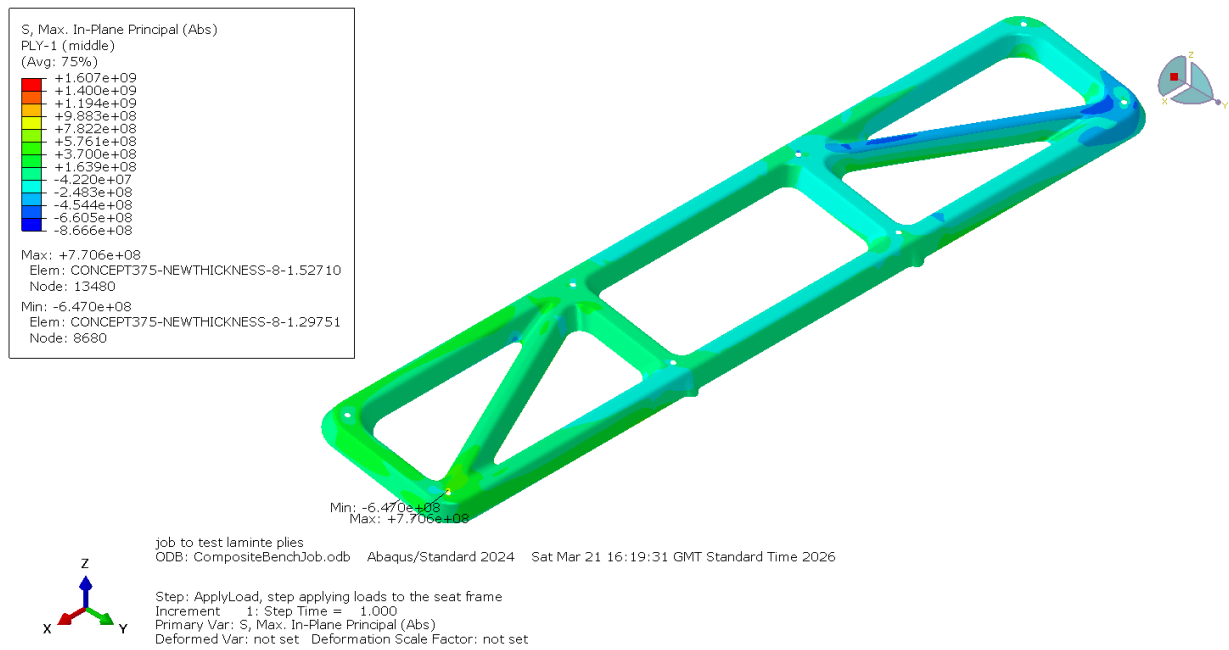


Figure E.1: Stress contour of bench part ply 1

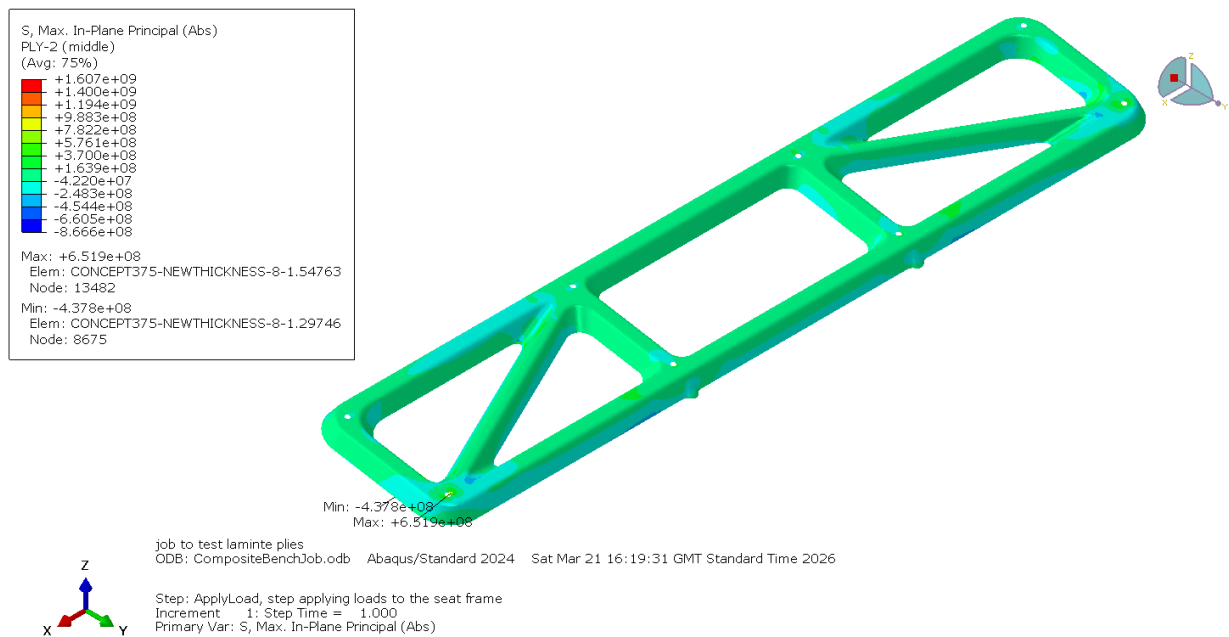


Figure E.2: Stress contour of bench part ply 2

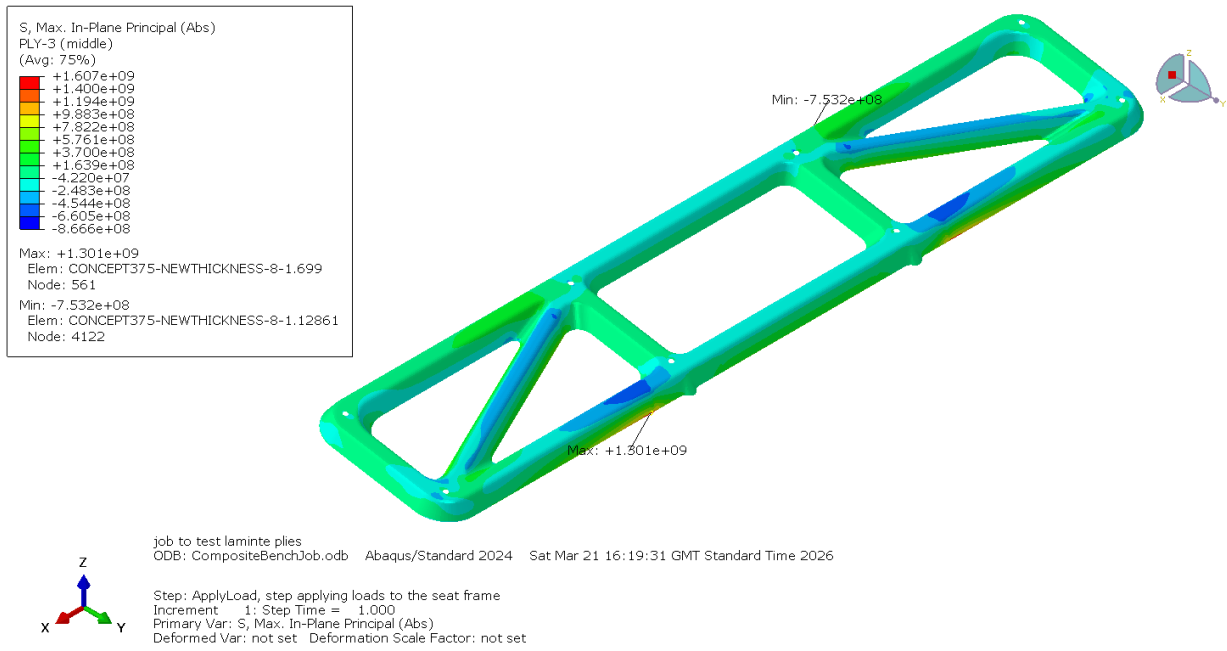


Figure E.3: Stress contour of bench part ply 3

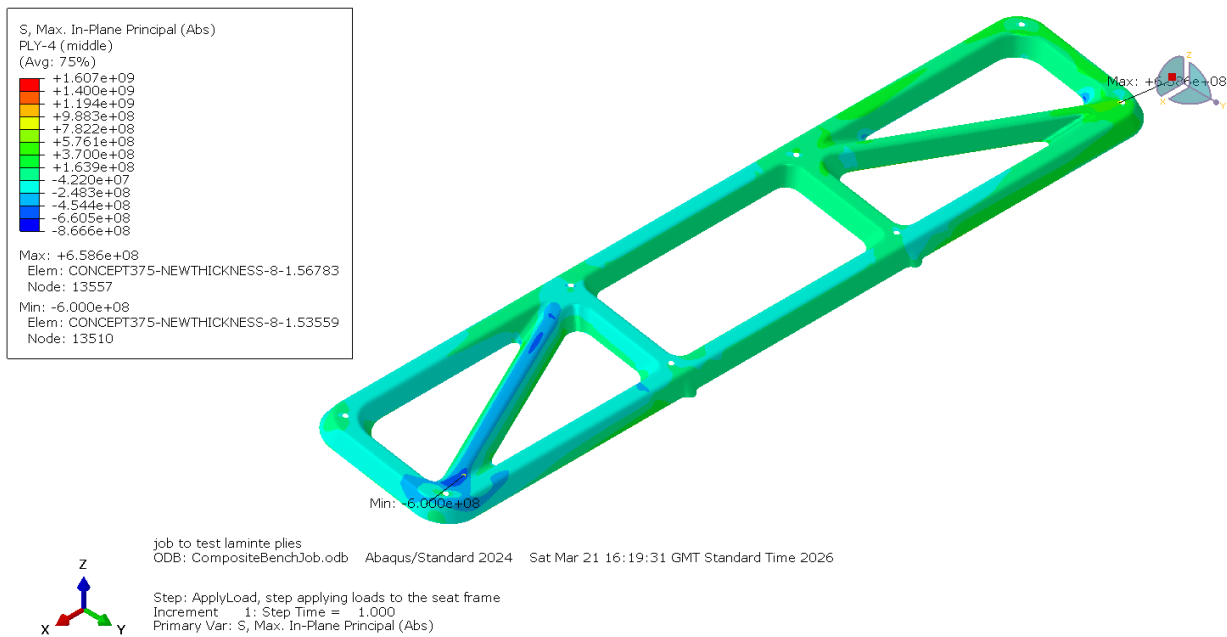


Figure E.4: Stress contour of bench part ply 4

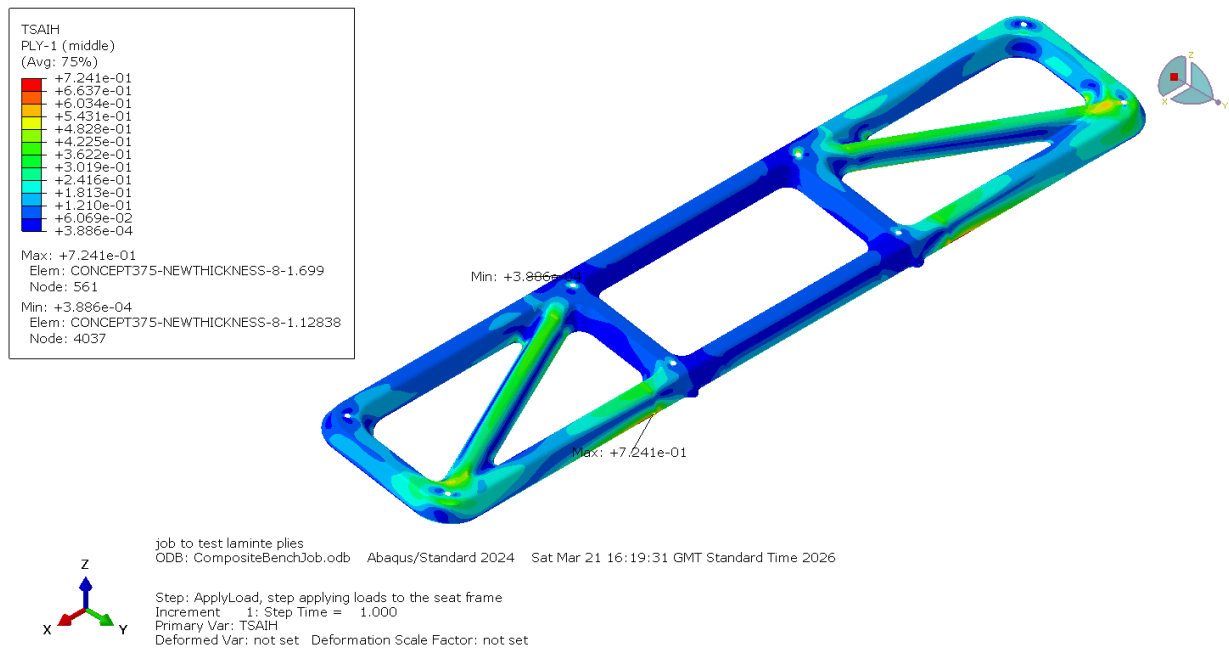


Figure E.5: Tsai-Hill contour of bench part ply 1

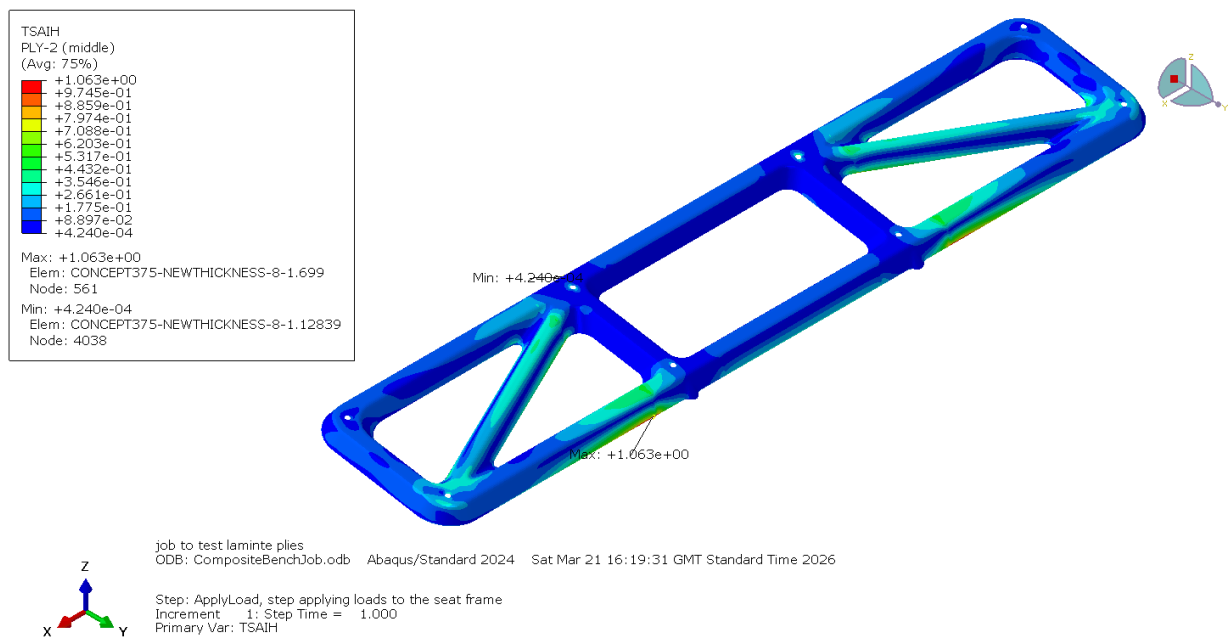


Figure E.6: Tsai-Hill contour of bench part ply 2

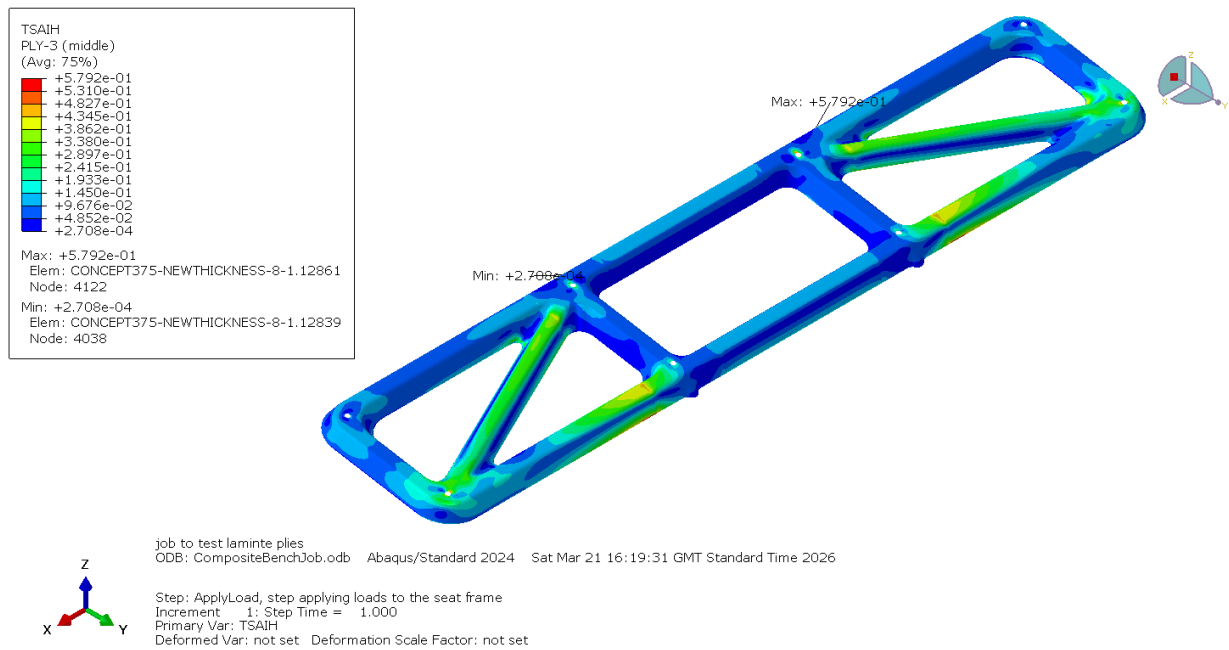


Figure E.7: Tsai-Hill contour of bench part ply 3

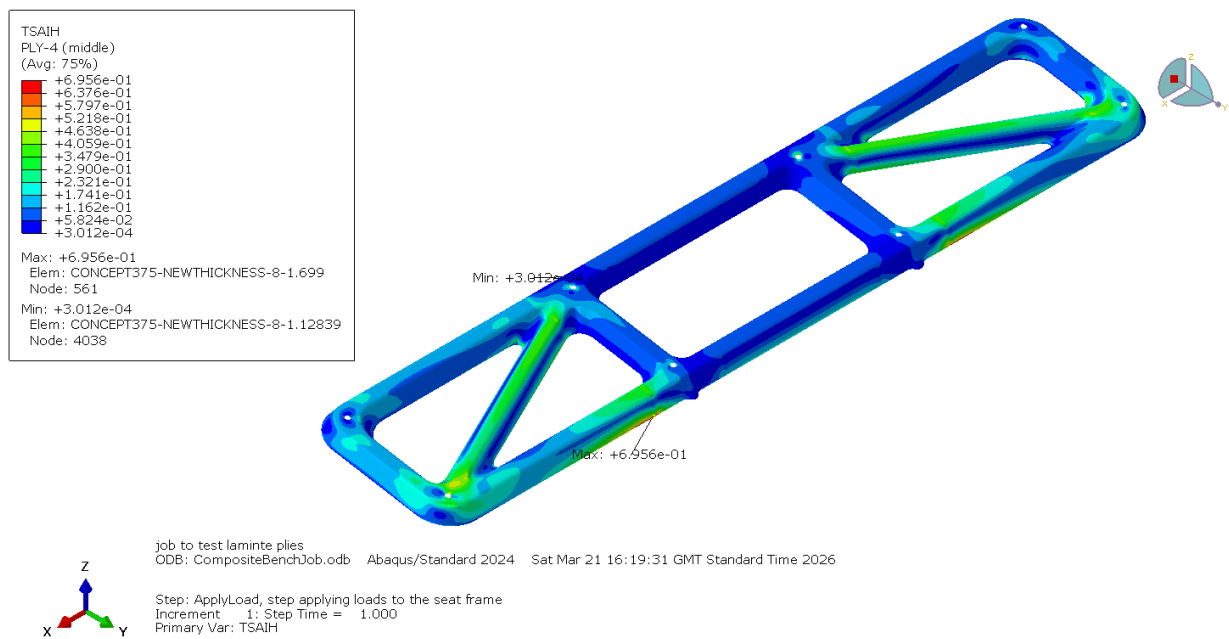
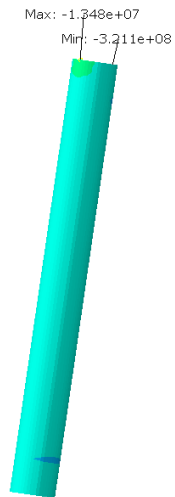
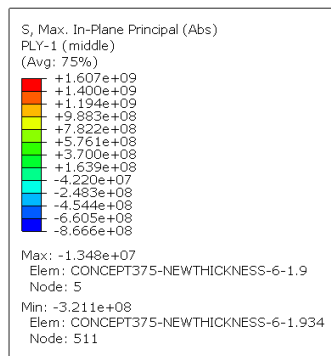


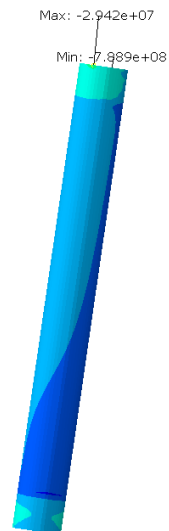
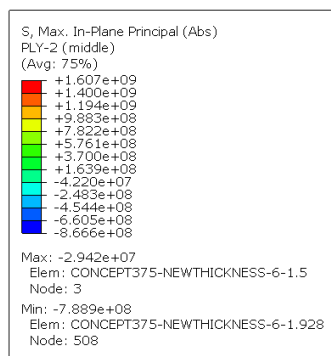
Figure E.8: Tsai-Hill contour of bench part ply 4



job to analyses front composite leg
ODB: CompositeFrontLegJob.odb Abaqus/Standard 2024 Sat Mar 21 15:40:10 GMT Standard Time 2026

Step: ApplyLoad, step applying loads to the seat frame
Increment 1: Step Time = 1.000
Primary Var: S, Max, In-Plane Principal (Abs)
Deformed Var: not set Deformation Scale Factor: not set

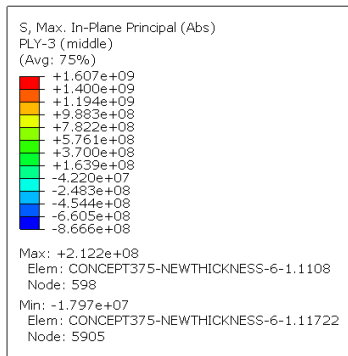
Figure E.9: Stress contour of front leg part ply 1



job to analyses front composite leg
ODB: CompositeFrontLegJob.odb Abaqus/Standard 2024 Sat Mar 21 15:40:10 GMT Standard Time 2026

Step: ApplyLoad, step applying loads to the seat frame
Increment 1: Step Time = 1.000
Primary Var: S, Max, In-Plane Principal (Abs)

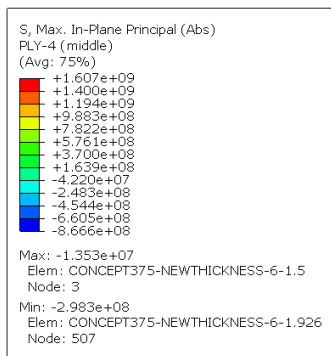
Figure E.10: Stress contour of front leg part ply 2



job to analyses front composite leg
ODB: CompositeFrontLegJob.odb Abaqus/Standard 2024 Sat Mar 21 15:40:10 GMT Standard Time 2026

Step: ApplyLoad, step applying loads to the seat frame
Increment 1: Step Time = 1.000
Primary Var: S, Max, In-Plane Principal (Abs)
Deformed Var: not set Deformation Scale Factor: not set

Figure E.11: Stress contour of front leg part ply 3



job to analyses front composite leg
ODB: CompositeFrontLegJob.odb Abaqus/Standard 2024 Sat Mar 21 15:40:10 GMT Standard Time 2026

Step: ApplyLoad, step applying loads to the seat frame
Increment 1: Step Time = 1.000
Primary Var: S, Max, In-Plane Principal (Abs)

Figure E.12: Stress contour of front leg part ply 4

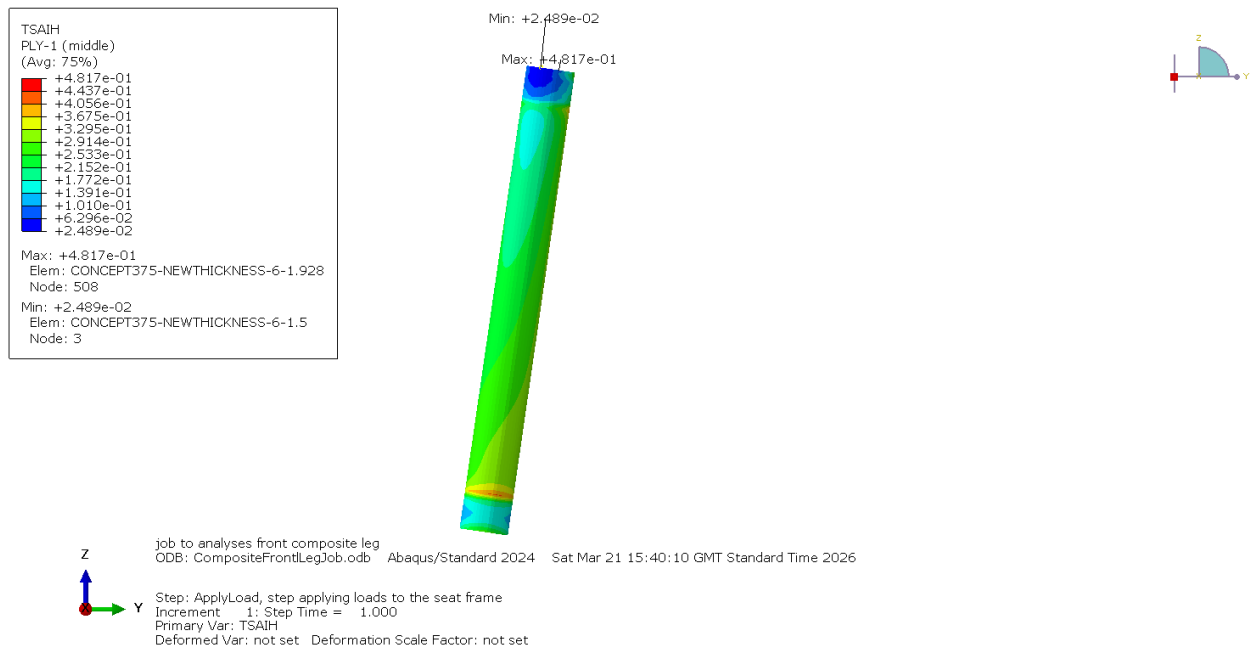


Figure E.13: Tsai-Hill contour of front leg part ply 1

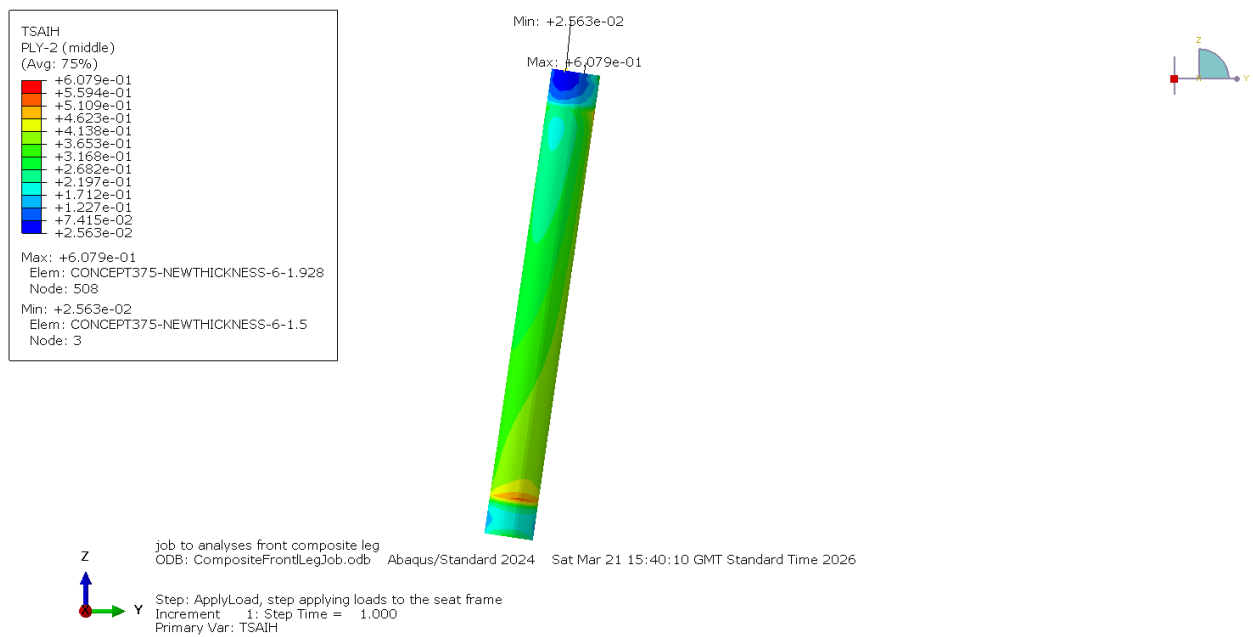


Figure E.14: Tsai-Hill contour of front leg part ply 2

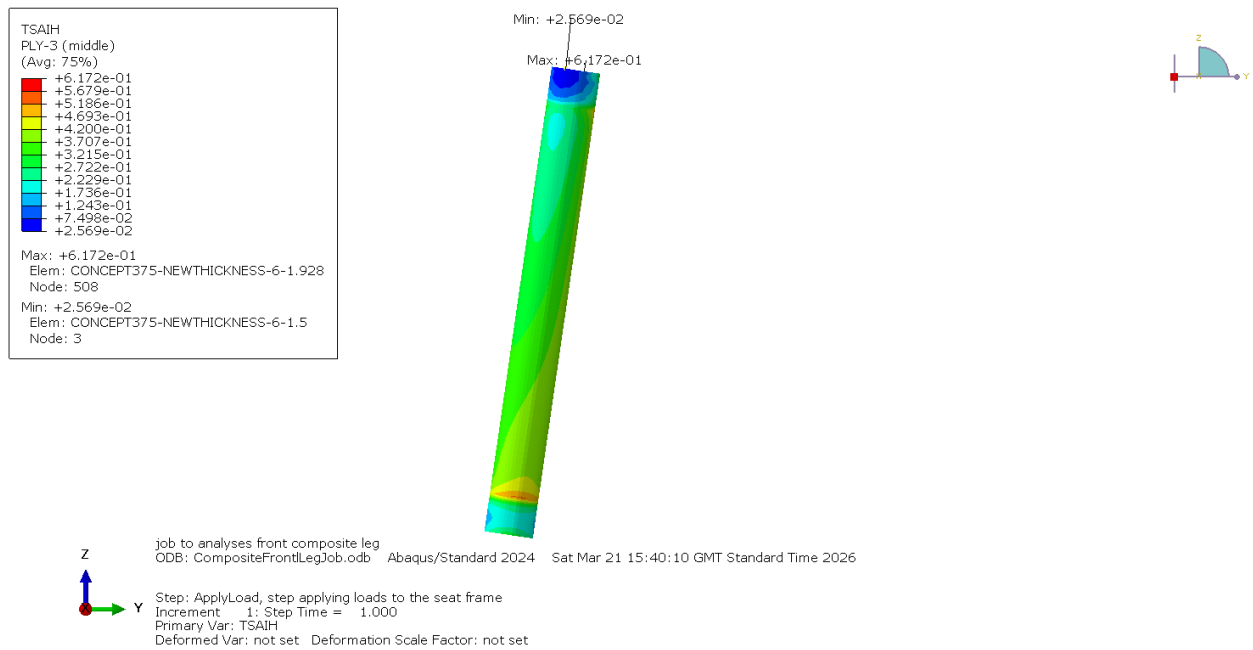


Figure E.15: Tsai-Hill contour of front leg part ply 3

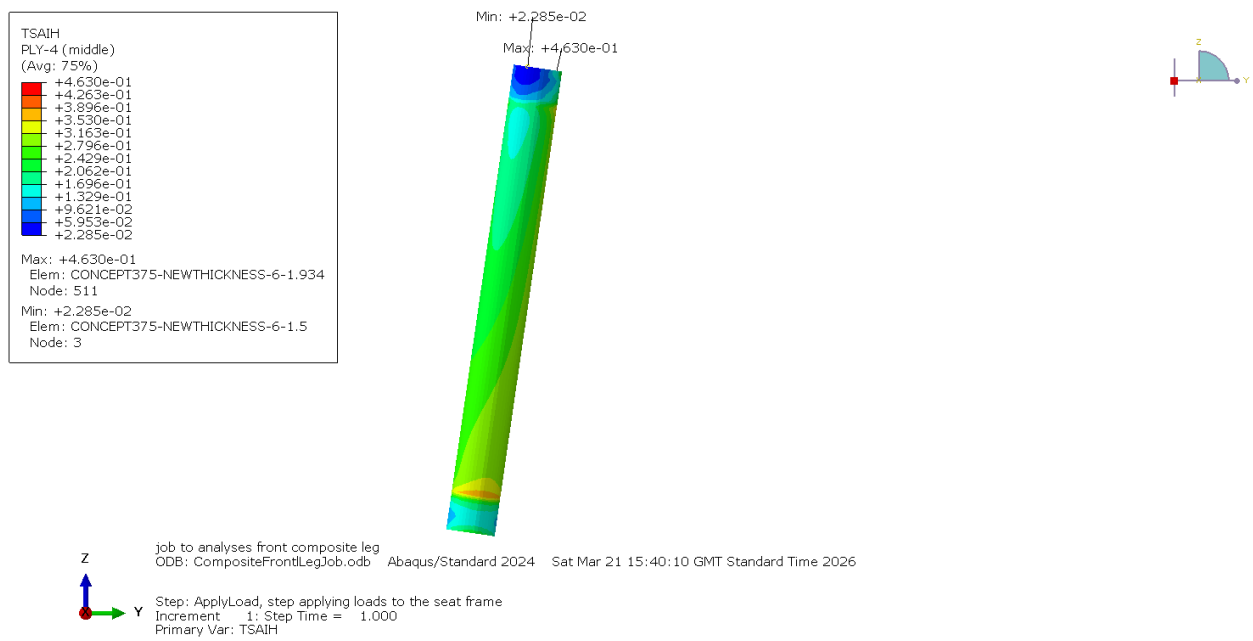


Figure E.16: Tsai-Hill contour of front leg part ply 4

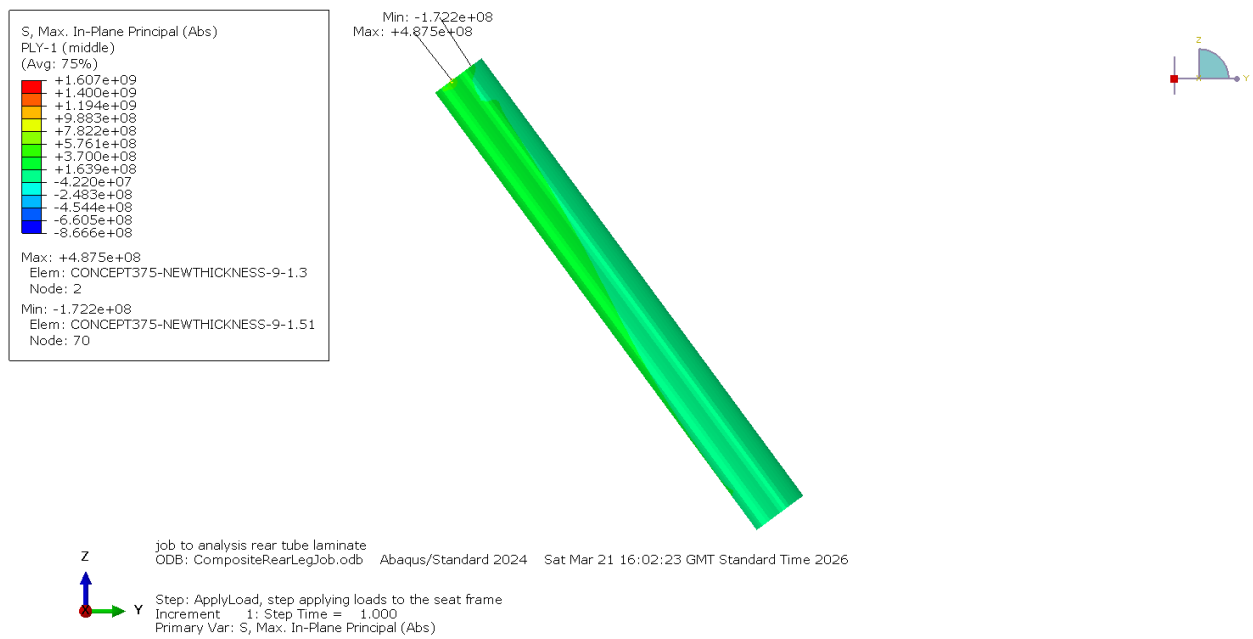


Figure E.17: Stress contour of rear leg part ply 1

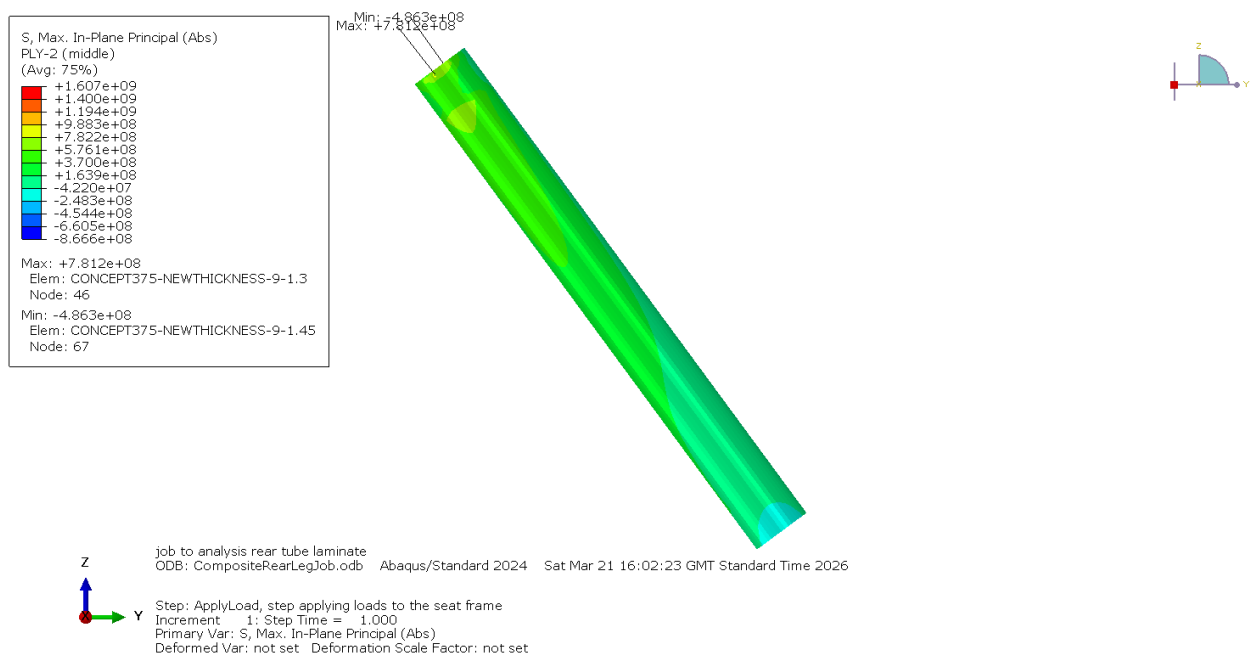


Figure E.18: Stress contour of rear leg part ply 2

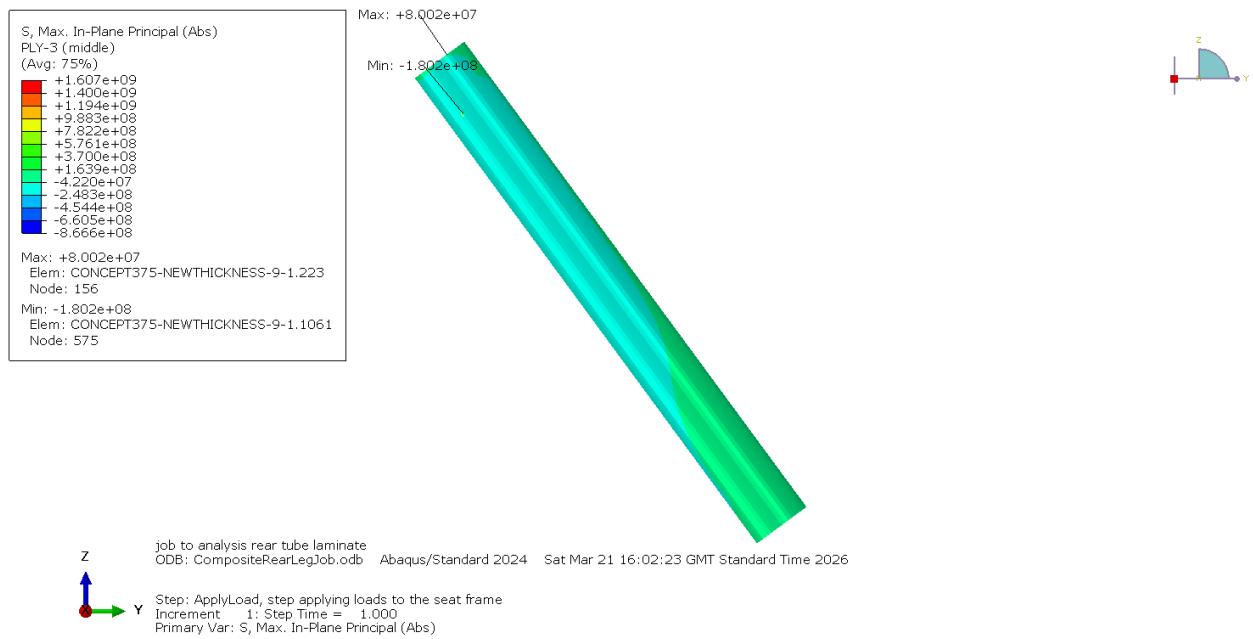


Figure E.19: Stress contour of rear leg part ply 3

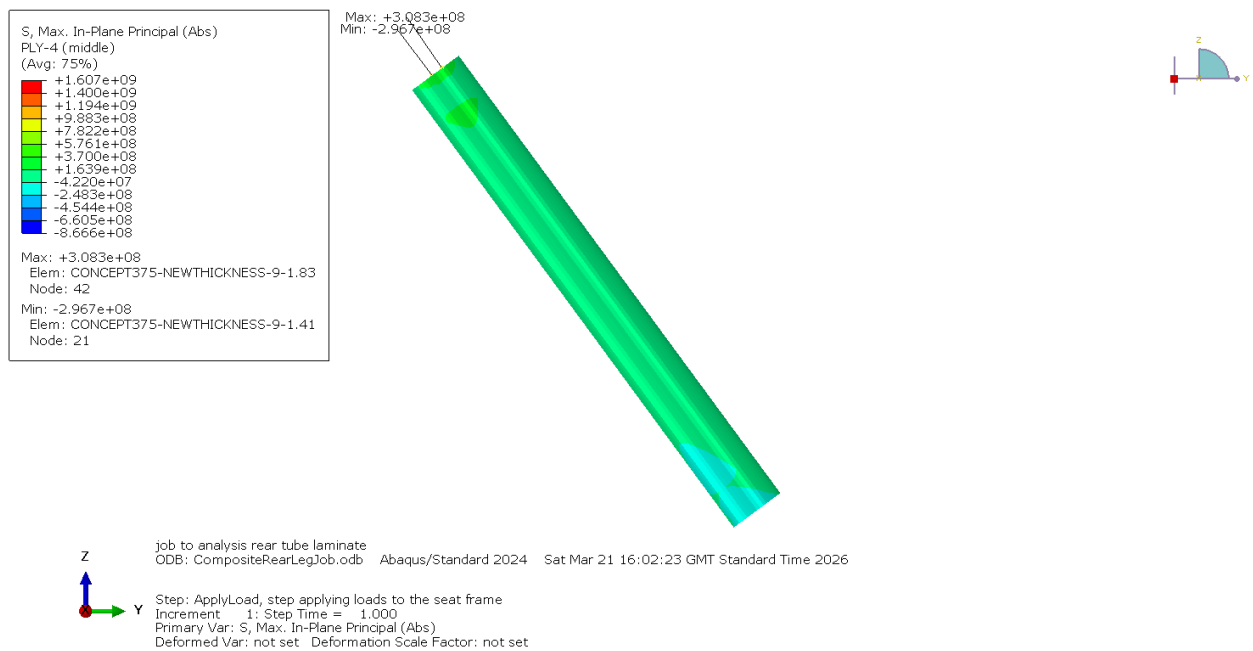


Figure E.20: Stress contour of rear leg part ply 4

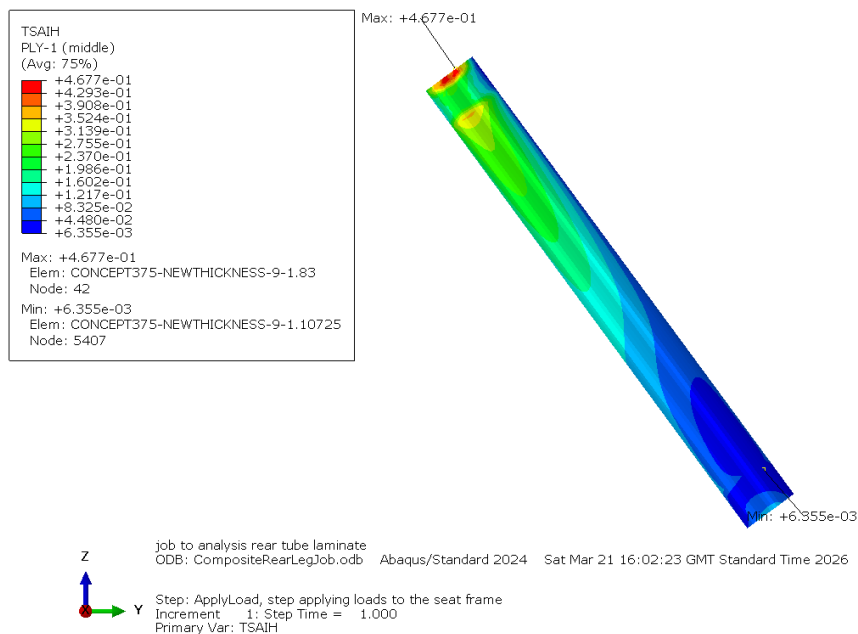


Figure E.21: Tsai-Hill contour of rear leg part ply 1

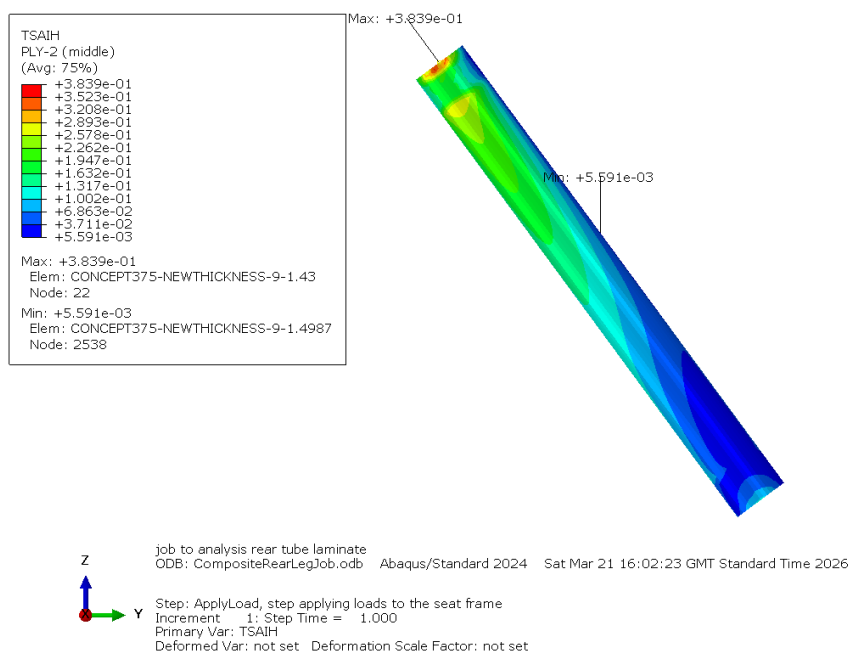


Figure E.22: Tsai-Hill contour of rear leg part ply 2

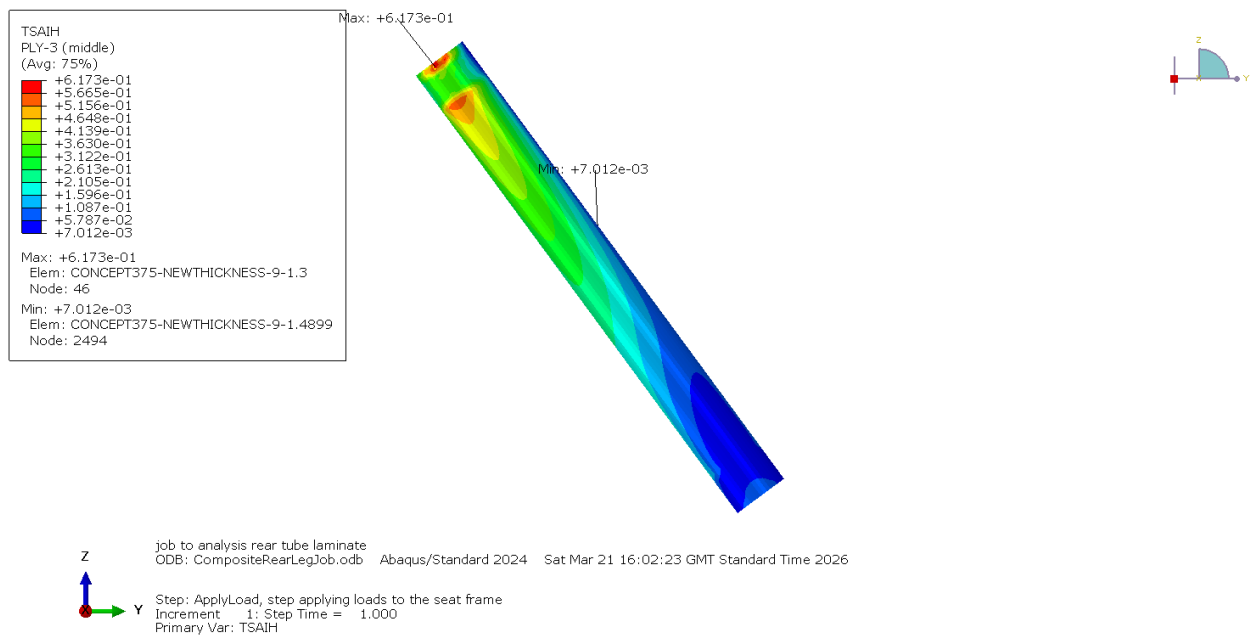


Figure E.23: Tsai-Hill contour of rear leg part ply 3

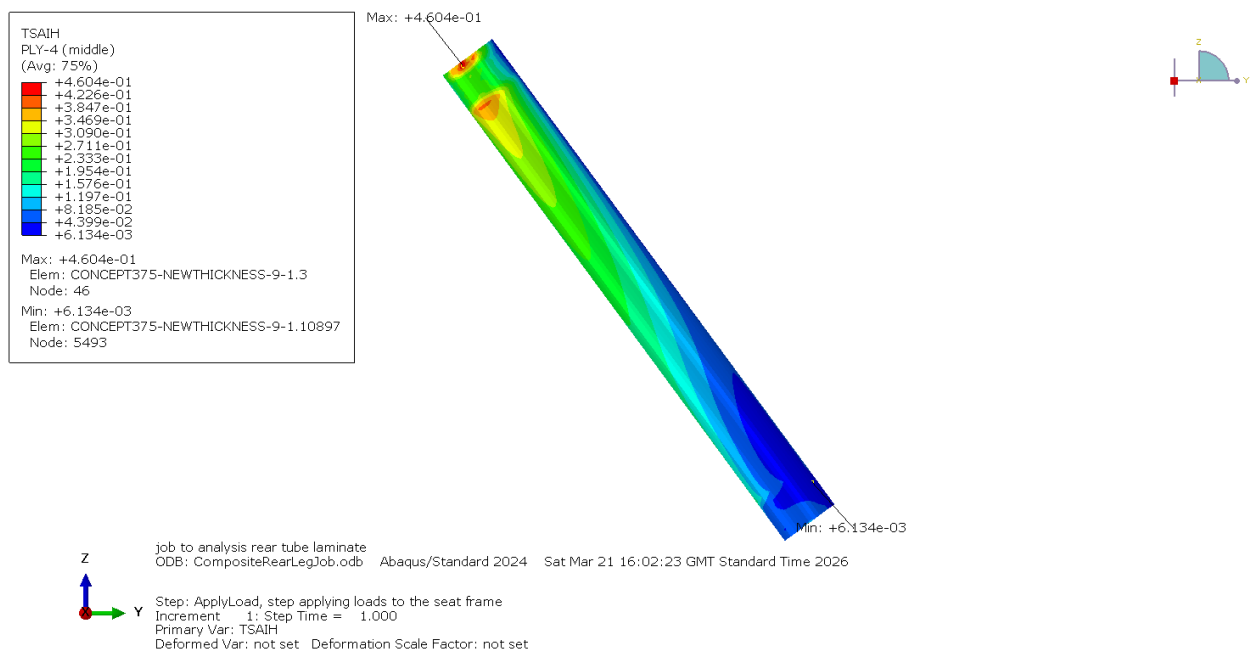


Figure E.24: Tsai-Hill contour of rear leg part ply 4